CALIBTRACK: LABORATORY EQUIPMENT CALIBRATION & CERTIFICATION MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

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BONAFIDE CERTIFICATE

Certified that this project report titled "**CALIBTRACK: LABORATORY EQUIPMENT CALIBRATION & CERTIFICATION MANAGEMENT SYSTEM**" is the bonafide work of **DHARSHINI G 240701121,** who carried out the individual project work on the Zoho Creator low-code platform for the course **CB23F35 – Low-Code Application Development using Zoho Creator**, during the academic year **[2026–2027], Semester [V],** under my supervision.

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held on [05-10-2026].

INTERNAL EXAMINER	EXTERNAL EXAMINER

DECLARATION

I, DHARSHINI G 240701121 hereby declare that the project report titled **"CALIBTRACK: LABORATORY EQUIPMENT CALIBRATION & CERTIFICATE MANAGEMENT SYSTEM",** submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me. I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

Date: [05-10-2026]

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DHARSHINI G

ABSTRACT

Laboratories use measuring instruments and other equipment that require regular calibration, certification, inspection, and maintenance. Managing these activities through separate spreadsheets and paper records can make it difficult to track schedules, access past results, monitor certificates, and assign technicians. This may lead to delays, missed activities, and additional administrative work. CALIBTRACK was developed to manage these activities through a centralised low-code application.

The application is developed using Zoho Creator and includes eight forms: Laboratory, Equipment, Technician, Calibration Schedule, Calibration Record, Calibration Certificate, Maintenance Request, and Equipment Inspection. It also includes reports and a dashboard displaying equipment details, calibration status, and maintenance information. Deluge scripts and workflows automate activities such as technician assignment, failed-calibration alerts, and email notifications. The Calibration Record form also contains a Calibration Parameters subform for recording individual measurements.

The application was tested using sample records, and key notifications, including technician assignment and failed-calibration alerts, were verified. The dashboard displays summary figures and charts based on the available records. These features help organise laboratory equipment information, reduce manual tracking, and improve the monitoring of calibration and maintenance activities. CALIBTRACK demonstrates how Zoho Creator can support an efficient and centralised laboratory management process.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Calibration Management; Laboratory Equipment.

PROJECT AT A GLANCE

Attribute	Details
Project Title	CALIBTRACK – Laboratory Equipment Calibration & Certification Management System
Application Domain	Laboratory Equipment Management
Student Name & Register No.	DHARSHINI G 240701121
Zoho Creator Edition	Education
Application Name (Link Name)	CALIBTRACK
Published / Live URL	https://creatorapp.zoho.in/dharshini.g2.2024.cse_raj205/calibtrack/#Page:Calibration_Dashboard
Demo Credentials (per role)	Not Applicable – Role-based demo users have not yet been configured.
Number of Forms / Subforms	8 Forms / 1 Subform
Number of Reports	List: 15; Kanban: 0; Calendar: 0; Pivot/Chart: 0; Others: 0
Number of Pages / Dashboards	1 Dashboard
Number of Workflows	Form: 10; Schedule: 1
Deluge Functions / Custom Actions	Deluge scripts used in form workflows and scheduled automation
Integrations	Zoho Creator native email notifications and automation
Roles & Profiles Defined	CEO, Laboratory Technician, Equipment Manager, Calibration Supervisor, Read-only Auditor
Environments Used	Development
Date of Production Deployment	Not Yet Deployed

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
ER	Entity–Relationship
FR	Functional Requirement
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UI	User Interface
UAT	User Acceptance Testing
Zoho Creator	Low-code application development platform
Deluge	Zoho scripting language
KPI	Key Performance Indicator

CHAPTER 1

INTRODUCTION

1.1 Background

Laboratories play an important role in educational institutions, research centres and testing organisations. They are used for practical sessions, experiments, research activities and different types of testing. These laboratories contain equipment such as measuring instruments, testing devices and other specialised tools. To obtain reliable results, the equipment must be maintained in good working condition and calibrated at regular intervals.

Calibration is the process of checking an instrument's readings against a known standard to determine whether it is providing accurate measurements. Regular calibration helps identify measurement errors and ensures that equipment is suitable for its intended use. Calibration details, results and certificates must be recorded properly so that the laboratory can refer to them whenever required. Equipment may also need periodic inspection and maintenance to prevent problems during its use.

This project considers a typical college laboratory that handles multiple equipment items used by students, faculty members and laboratory staff. The laboratory may have different categories of equipment, each with its own identification details, calibration requirements and maintenance schedule. The number of equipment items and related records can increase over time as new equipment is purchased and existing equipment is serviced or calibrated.

In a laboratory, several people are involved in managing equipment and calibration activities. Laboratory staff maintain equipment details and monitor its availability. Technicians carry out calibration activities, record the readings and enter the results. Equipment managers monitor equipment condition and coordinate maintenance

requirements. Supervisors oversee calibration activities and check whether the necessary records and certificates are maintained. Each person has a different responsibility, but all of them depend on accurate and updated equipment information.

In the existing process, equipment information is often maintained using registers, spreadsheets or separate documents. Details such as equipment name, identification number, category, laboratory location and current status are recorded manually. When equipment is sent for calibration, the calibration date, results and next calibration date are also entered into records. Certificates received after calibration may be stored separately as printed documents or digital files.

Tracking calibration schedules is another important activity. Laboratory staff need to identify which equipment is due for calibration, which items have upcoming calibration dates and which items have already passed their due dates. When this information is maintained manually, staff must regularly check the records to identify equipment that requires attention. If the records are not updated on time, it may become difficult to monitor the calibration status of all equipment.

Calibration results also need to be maintained for future reference. During calibration, technicians record the required parameters and observations. The results help determine whether the equipment has passed or failed the calibration process. Previous calibration records are useful when reviewing the performance of equipment over time. However, when these records are stored in different files, finding the history of a particular equipment item may take additional time.

Apart from calibration, equipment inspection and maintenance are also necessary. Equipment may develop faults or require servicing because of regular usage. In such situations, a maintenance request must be recorded and followed up until the issue is addressed. Inspection details are also important because they help identify equipment problems and maintain proper records of its condition. Without a common system, tracking these activities along with calibration records can become difficult.

1.2 Problem Statement

Maintaining laboratory equipment and calibration details manually can take time and make it difficult to keep records organised. When information is stored in different registers and files, staff may find it difficult to check equipment details, track calibration dates and identify pending maintenance activities. This can affect the timely monitoring of laboratory equipment.

The main problems identified are:

1. Difficulty in maintaining records: Equipment, laboratory and technician details are stored separately, making it difficult to find and update information.
2. Difficulty in tracking calibration schedules: Staff may find it difficult to remember upcoming and overdue calibration dates when schedules are maintained manually.
3. Lack of proper calibration history: Previous calibration results and related details may be difficult to access when records are scattered across different files.
4. Delay in maintenance tracking: Pending maintenance requests and equipment inspection issues may not be followed up on time.
5. Time-consuming report preparation: Preparing equipment, calibration and maintenance reports manually requires additional time and effort.

1.3 Objectives

The main objectives of the CALIBTRACK project are:

- To design and develop a centralised application for managing laboratory equipment and calibration activities.

- To create forms for maintaining laboratory, equipment, technician, calibration schedule, calibration record, certificate, maintenance request and equipment inspection details.
- To automate selected activities using Zoho Creator workflows and Deluge scripts, including schedule status updates and notifications.
- To generate reports for viewing equipment details, calibration history, certificate expiry and maintenance activities.
- To develop a dashboard that displays important equipment, calibration and maintenance information in one place.
- To test the implemented forms, reports, workflows and dashboard to check whether the application works as expected.

1.4 Scope of the Project

The project focuses on developing a laboratory equipment calibration and certification management application using Zoho Creator.

In-Scope

- Maintaining laboratory, equipment and technician details.
- Creating and managing calibration schedules and records.
- Recording calibration parameters and results.
- Maintaining calibration certificate details.
- Recording maintenance requests and equipment inspections.
- Using selected workflows and Deluge scripts for automation.
- Generating reports to view equipment, calibration and maintenance information.
- Displaying important information through the Calibration Dashboard.
- Testing the implemented application features.

Out-of-Scope

- Direct connection with calibration instruments or laboratory hardware.
- Automatic measurement of equipment accuracy.
- Integration with external laboratory management systems.
- Integration with third-party applications through external APIs.
- Full production deployment in a real laboratory environment.

The application is developed mainly for laboratory staff, technicians, equipment managers and supervisors. It is intended to be used through the Zoho Creator platform.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected for this project because it provides tools to develop applications with less manual coding. Forms can be created to collect information, reports can be used to display stored data, and workflows can be configured to automate selected activities.

Using Zoho Creator made it easier to develop the different modules of CALIBTRACK within the project period. Deluge scripts were used wherever additional logic was required. The platform also provides a dashboard to display important information in a single place.

Another reason for selecting Zoho Creator is that changes can be made to forms, reports and workflows without developing the entire application again. This makes it suitable for a project where requirements may change during development. Therefore, Zoho Creator was a suitable platform for developing CALIBTRACK as a low-code application.

Reference: Zoho Corporation. (n.d.). Creating an Application – Zoho Creator FAQs. Zoho Creator Help.

1.6 Organisation of the Report

This report is organised into nine chapters. Chapter 1 introduces the project, explains the background, problem statement, objectives, scope and reason for selecting the low-code approach. Chapter 2 presents the study of the existing system and the proposed system, along with the requirements and feasibility of the project. Chapter 3 explains the system analysis and design, including the application structure and database design. Chapter 4 describes the development environment and tools used for building the application.

Chapter 5 explains the implementation of the CALIBTRACK application, including its forms, reports, workflows and dashboard. Chapter 6 discusses the testing carried out on the application and the results obtained. Chapter 7 presents the screenshots and working details of the developed application. Chapter 8 discusses the results, benefits and limitations of the project. Finally, Chapter 9 provides the conclusion and possible future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system, laboratory equipment details are maintained using manual registers and spreadsheets. Staff record equipment information and monitor calibration schedules by checking the records regularly. They contact technicians to arrange calibration and enter the results after the process is completed. Calibration certificates are stored separately for future reference. If equipment fails calibration or develops a fault, maintenance requests are recorded and followed up manually. Since all these activities are handled separately, maintaining records, finding information and tracking equipment status can become difficult and time-consuming.

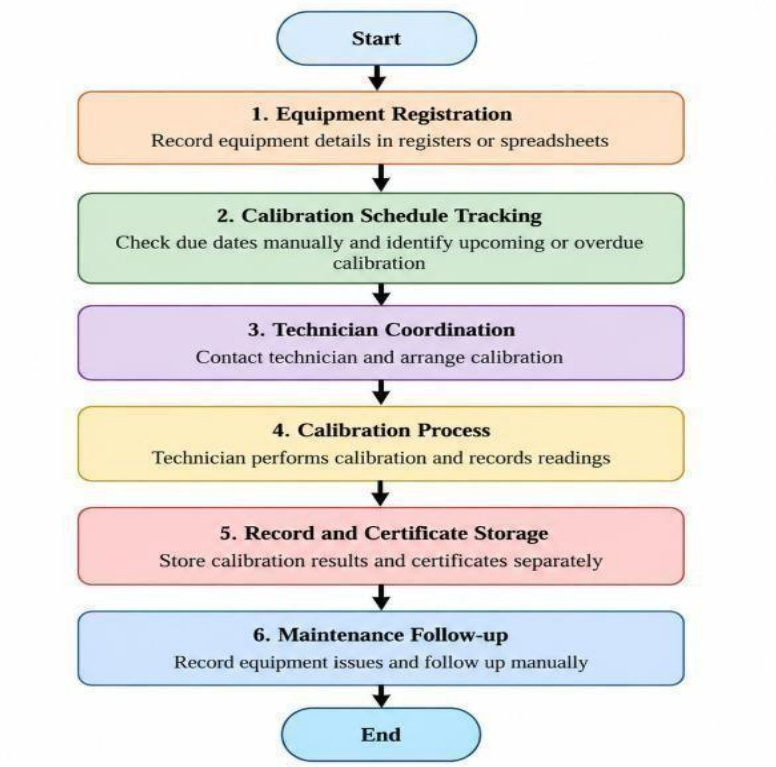


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Equipment and calibration records are maintained separately, making information difficult to access.
- Calibration schedules are tracked manually, which may lead to missed or delayed calibration.
- Maintenance activities and certificate records are difficult to monitor and update regularly.

2.3 Low-Code versus Traditional Development

Criterion	Traditional Development	Low-Code on Zoho Creator
Development time	Requires more time for coding, testing and deployment.	Visual tools reduce development time.
Skills required	Requires programming, database and web development skills.	Requires form design, platform configuration and Deluge knowledge.
Hosting and infrastructure	Hosting, database and server setup must be managed separately.	Hosting and platform infrastructure are managed by Zoho Creator.
Mobile access	May require separate mobile application development.	Provides mobile access options through the platform.
Security and access control	Managed through custom coding and server configuration.	Managed using users, roles, profiles and permissions.
Maintenance	Changes usually require code modifications and testing.	Forms, reports and workflows can be modified through the platform.
Customisation limits	Offers greater flexibility through custom coding.	Customisation depends on platform features and Deluge scripting.
Cost	Depends on development, hosting and maintenance expenses.	Depends on the Zoho Creator plan and development requirements.

Table 2.1: Low-Code and Traditional Development Comparison

2.4 Literature Survey

S.No.	Author / Source	Title / Product	Key Focus	Relevance to CALIBTRACK
1	Zoho Corporation	Zoho Creator Documentation	Explains forms, reports, workflows and application development.	Supports the development of CALIBTRACK modules.
2	Zoho Corporation	Deluge scripting documentation	Explains scripting, functions and automation in Zoho Creator.	Supports the automation scripts used in CALIBTRACK.
3	Gartner (2021)	Magic Quadrant for Enterprise Low-Code Application Platforms	Discusses low-code platforms and their role in application development.	Provides background on the use of low-code platforms.
4	ISO (2017)	ISO/IEC 17025:2017 – General Requirements for the Competence of Testing and Calibration Laboratories	Describes requirements related to testing and calibration laboratory competence.	Provides a quality and calibration management reference for the project.
5	Mendling et al. (2021)	Low-Code Development Platforms: A Literature Review	Discusses low-code development concepts and their benefits and limitations.	Supports the selection of low-code development for CALIBTRACK.

Table 2.2: Summary of Literature Survey

2.5 Emerging Trends in Low-Code Development

Low-code development is changing with the introduction of new technologies and development methods. The following trends are relevant to the CALIBTRACK application.

1. AI-Assisted Application Building: AI tools help developers create forms, generate suggestions and build application components using simple instructions. During the

development of CALIBTRACK, Zia was used to assist with dashboard changes and report arrangements.

2. Citizen Development: Low-code platforms allow users with basic technical knowledge to develop applications using visual tools. Zoho Creator helped in building CALIBTRACK forms, reports and workflows without writing code for every feature.

3. Hyper-Automation: Hyper-automation connects multiple activities through workflows and automation. In CALIBTRACK, selected workflows and Deluge scripts are used for schedule status updates and notifications, reducing some manual work.

4. Composable Applications: This approach involves building an application using separate modules that work together. CALIBTRACK follows this approach by using individual forms for equipment, laboratories, technicians, calibration records, certificates, maintenance requests and inspections. These modules are connected to support the overall process.

These trends influenced the design of CALIBTRACK by encouraging the use of visual development, selected automation, reusable application components and a centralised dashboard. They helped make the application easier to develop and manage using Zoho Creator.

2.6 Proposed System

CALIBTRACK is developed using Zoho Creator to manage laboratory equipment and calibration activities in one application. It provides forms to maintain laboratory, equipment and technician details. It also manages calibration schedules, calibration records, certificates, maintenance requests and equipment inspections. Reports help users access the stored information easily. The dashboard displays important equipment, calibration and maintenance details. Selected workflows and Deluge scripts support schedule updates and notifications. Overall, the proposed system reduces manual work and keeps laboratory records organised.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Responsibilities
Laboratory Technician	Person who handles calibration activities.	Update calibration records, results and equipment details.
Equipment Manager	Person responsible for equipment management.	Maintain equipment information and monitor maintenance requests.
Calibration Supervisor	Person who monitors calibration activities.	Monitor calibration schedules, results and certificates.
Read-only Auditor	User who views application records.	View reports and records for reference.

Table 3.1: Stakeholders and User Roles

3.2 Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow users to add and maintain laboratory details.	Laboratory Technician	High
FR-02	The system shall allow users to register equipment and update equipment details.	Equipment Manager	High
FR-03	The system shall allow users to add and maintain technician details.	Equipment Manager	High

FR-04	The system shall allow users to create calibration schedules and assign technicians.	Calibration Supervisor	High
FR-05	The system shall allow users to record calibration results and parameter details.	Laboratory Technician	High
FR-06	The system shall allow users to maintain calibration certificate details.	Laboratory Technician	Medium
FR-07	The system shall allow users to record maintenance requests and equipment inspection details.	Equipment Manager	Medium
FR-08	The system shall display equipment, calibration and maintenance information through reports and a dashboard.	Authorised users	High
FR-09	The system shall update calibration schedule status based on the calibration date.	Calibration Supervisor	Medium
FR-10	The system shall send configured email notifications for selected events, such as technician assignment and failed calibration.	Relevant Users	Medium

Table 3.2: Functional Requirements

3.3 Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	The application shall provide clear forms, labels, navigation and reports for easy use.
NFR-02	Data integrity	Mandatory fields and validations shall help prevent incomplete or incorrect data entry.
NFR-03	Security	Users shall access application data according to their configured roles and permissions.
NFR-04	Maintainability	Forms, reports and workflows shall have clear names and organised structures for easier maintenance.

NFR-05	Availability	The application shall be accessible through Zoho Creator when the platform and internet connection are available.
NFR-06	Auditability	Application records and changes shall be traceable using available platform features, where configured.

Table 3.3: Non-Functional Requirements

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator – Education Edition
Scripting	Deluge scripting language
Browser	Google Chrome / Microsoft Edge
Network	Internet connection required to access Zoho Creator.
Client Hardware	Desktop or laptop computer.
External services	Zoho Creator email notification functionality.

Table 3.4: Hardware and Software Requirements

3.5 Use Case Model

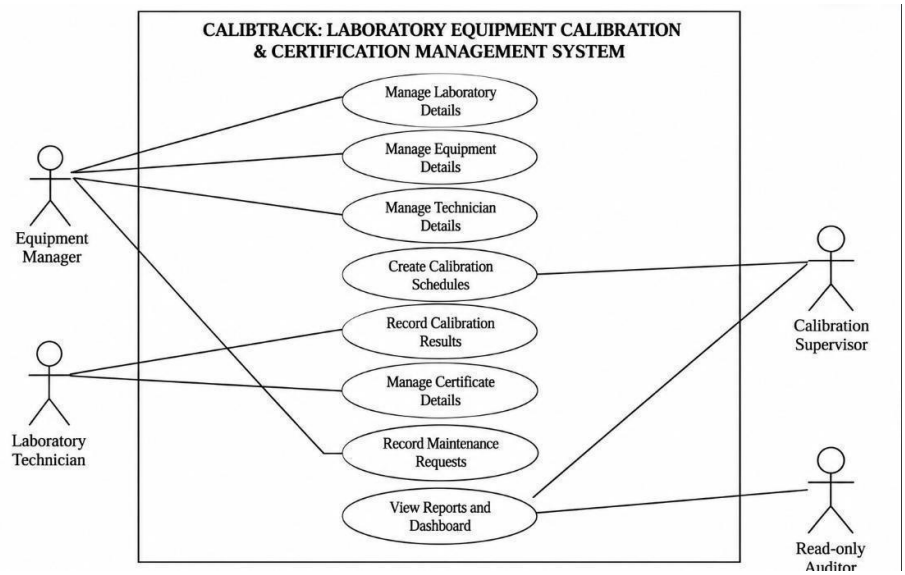


Figure 3.1: Use Case Diagram of CALIBTRACK

Field	Description
Use Case ID	UC-01
Use Case Name	Manage Equipment Details
Actor(s)	Equipment Manager
Pre-condition	User has access to the application.
Main Flow	1. User opens the Equipment form. 2. Enters equipment details. 3. Submits the form. 4. System saves the equipment record.
Alternate Flow	If mandatory details are missing, the form displays a validation message.
Post-condition	If mandatory details are missing, the form displays a validation message.
Related FR	FR-02

Table 3.5: Use Case Description – Manage Equipment Details

Field	Description
Use Case ID	UC-02
Use Case Name	Manage Calibration Records
Actor(s)	Laboratory Technician
Pre-condition	Equipment and calibration schedule details are available.
Main Flow	1. User opens the Calibration Record form. 2. Selects the equipment. 3. Enters calibration results and parameters. 4. Submits the record. 5. System saves the details.
Alternate Flow	If required fields are missing, the form displays a validation message.
Post-condition	Calibration results are stored for future reference.
Related FR	FR-05

Table 3.6: Use Case Description – Manage Calibration Records

Field	Description
Use Case ID	UC-03
Use Case Name	Manage Maintenance Requests
Actor(s)	Equipment Manager
Pre-condition	Equipment details are available in the application.
Main Flow	1. User opens the Maintenance Request form. 2. Selects the equipment. 3. Enters the issue details. 4. Submits the request. 5. System stores the maintenance request.
Alternate Flow	If mandatory details are missing, the form displays a validation message.
Post-condition	Maintenance request is recorded in the application.
Related FR	FR-07

Table 3.7: Use Case Description – Manage Maintenance Requests

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

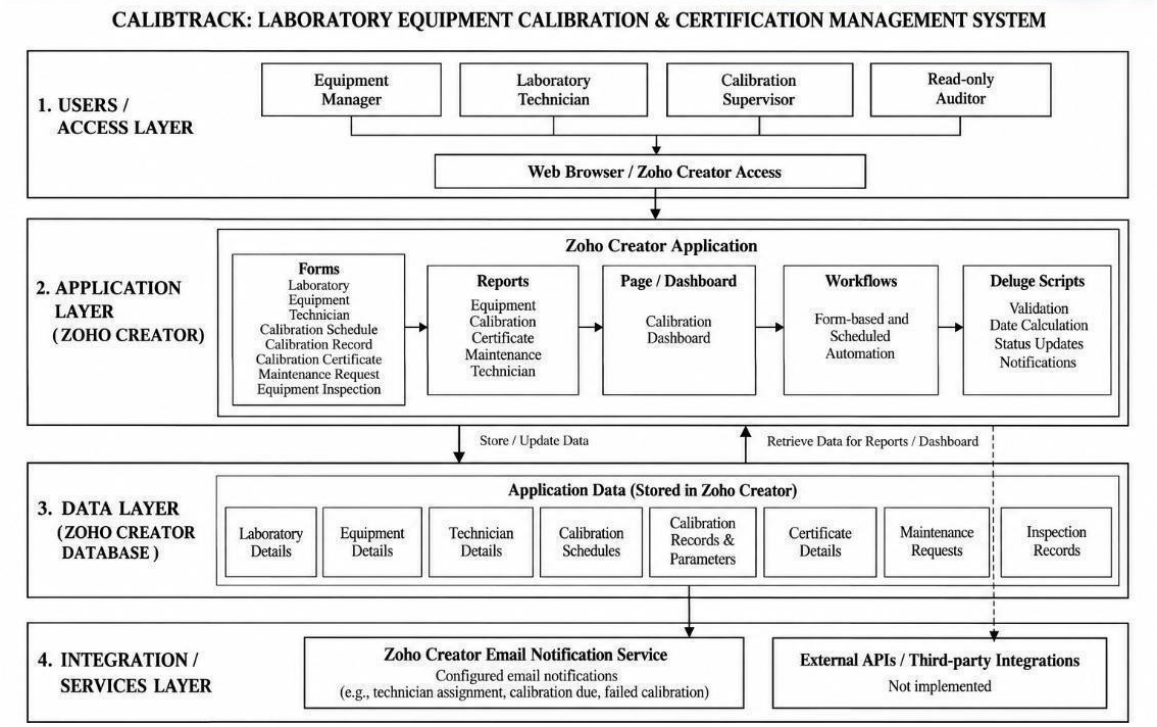


Figure 4.1: System Architecture of CALIBTRACK

The CALIBTRACK system architecture consists of four main layers.

1. Users / Access Layer: This layer includes the Equipment Manager, Laboratory Technician, Calibration Supervisor and Read-only Auditor. Users access the application through a web browser using Zoho Creator.

2. Application Layer: This layer contains the Zoho Creator application, including forms, reports, the Calibration Dashboard, workflows and Deluge scripts. These components are used to manage equipment details, calibration activities, certificates and maintenance records.

3. Data Layer: This layer stores all application information, including laboratory, equipment, technician, calibration schedule, calibration record, certificate, maintenance and inspection details.

4. Integration / Services Layer: This layer includes Zoho Creator email notification services used for configured notifications. External API integrations are not implemented in this project.

These four layers work together to provide a centralised system for managing laboratory equipment, calibration and certification activities.

4.2 Data Model

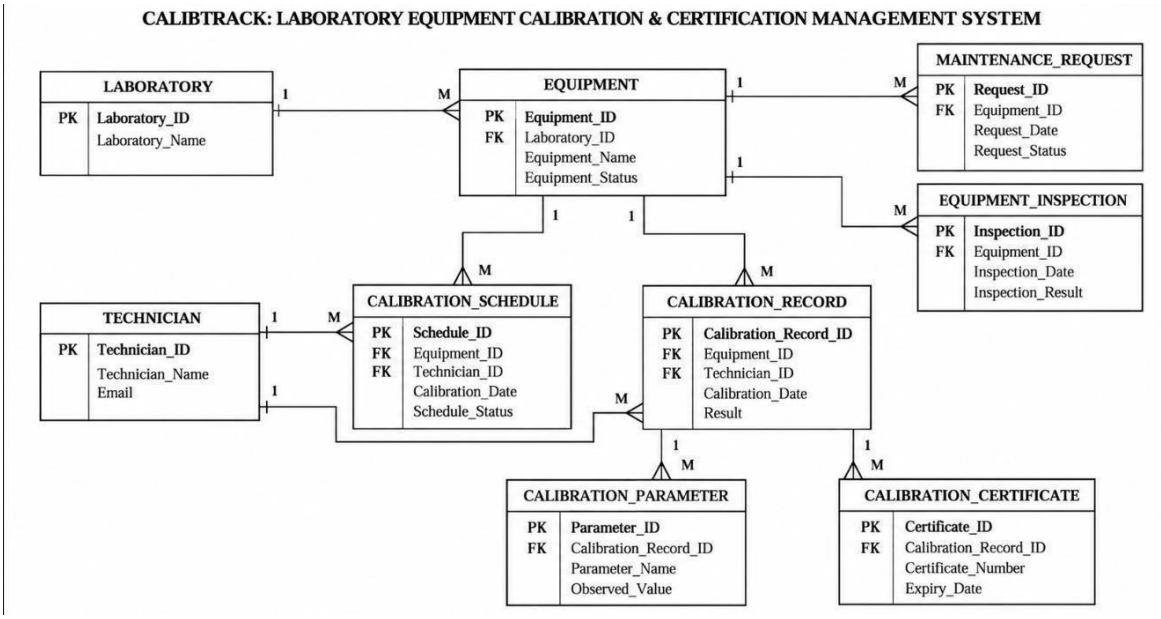


Figure 4.2: Entity–Relationship Diagram of CALIBTRACK

Entity (Form)	Key Fields	Related To	Relationship Type I	Implemented Using
Laboratory	Laboratory ID, Laboratory Name	Equipment	One-to-Many	Lookup in Equipment

Equipment	Equipment ID, Equipment Name, Equipment Status	Laboratory, Calibration Schedule, Calibration Record	One-to-Many	Lookup fields
Technician	Technician ID, Technician Name, Email	Calibration Schedule, Calibration Record	One-to-Many	Lookup fields
Calibration Schedule	Schedule ID, Calibration Date, Schedule Status	Equipment, Technician	Many-to-One	Lookup fields
Calibration Record	Record ID, Calibration Date, Result	Equipment, Technician, Calibration Parameters, Certificate	Parent–Child / One-to-Many	Lookup fields and Subform
Calibration Parameters	Parameter Name, Observed Value	Calibration Record	Parent–Child	Subform
Calibration Certificate	Certificate ID, Certificate Number, Expiry Date	Calibration Record	Many-to-One	Lookup field
Maintenance Request	Request ID, Request Date, Request Status	Equipment	Many-to-One	Lookup field
Equipment Inspection	Inspection ID, Inspection Date, Inspection Result	Equipment	Many-to-One	Lookup field

Table 4.1: Entities and Relationships

4.3 Form Design

Laboratory Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Laboratory ID	Laboratory_ID	Single Line	To confirm	To confirm	Laboratory identification
Laboratory Name	Laboratory_Name	Single Line	To confirm	To confirm	Name of laboratory
Department	Department	Drop Down	To confirm	Configured options	Department selection
Location	Location	Single Line	To confirm	To confirm	Laboratory location
Floor / Room No.	Floor_Room_No	Single Line	To confirm	To confirm	Room details
Lab In-charge	Lab_In_charge	Single Line	To confirm	To confirm	Responsible person
Contact Number	Contact_Number	Phone	To confirm	To confirm	Contact details
Email	Email	Email	To confirm	To confirm	Laboratory email
Laboratory Status	Laboratory_Status	Decision Box	To confirm	Configured options	Laboratory status
Description	Description	Multi Line	To confirm	Optional	Additional details

Table 4.2: Field Specification – Laboratory Form

Equipment Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Equipment ID	Equipment_ID	Auto Number	To confirm	System generated	Unique equipment ID
Equipment Name	Equipment_Name	Single Line	To confirm	—	Name of equipment
Equipment Category	Equipment_Category	Drop Down	To confirm	Configured options	Equipment category
Laboratory	Laboratory	Lookup	To confirm	Select laboratory	Associated laboratory
Manufacturer	Manufacturer	Single Line	To confirm	—	Manufacturer name
Model Number	Model_Number	Single Line	To confirm	—	Equipment model
Serial Number	Serial_Number	Single Line	To confirm	—	Serial number
Purchase Date	Purchase_Date	Date	To confirm	—	Date of purchase
Purchase Cost	Purchase_Cost	Currency	To confirm	—	Equipment cost
Last Calibration Date	Last_Calibration_Date	Date	To confirm	—	Previous calibration date
Calibration Interval	Calibration_Interval	Drop Down	To confirm	Configured options	Calibration frequency
Next Calibration Date	Next_Calibration_Date	Date	To confirm	—	Next due date
Equipment Status	Equipment_Status	Decision Box	To confirm	Configured options	Current equipment statu
Equipment Image	Equipment_Image	Image	To confirm	—	Equipment photograph
Equipment Description	Equipment_Description	Multi Line	To confirm	Optional	Additional details

Table 4.3: Field Specification – Equipment Form

Technician Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Technician ID	Technician_ID	Auto Number	To confirm	System generated	Unique technician ID
Technician Name	Technician_Name	Single Line	To confirm	—	Technician name
Specialization	Specialization	Drop Down	To confirm	Configured options	Area of expertise
Phone Number	Phone_Number	Phone	To confirm	—	Contact number
Email	Email	Email	To confirm	—	Email address
Qualification	Qualification	Single Line	To confirm	—	Educational qualification
Experience	Experience	Number	To confirm	—	Years of experience
Technician Status	Technician_Status	Decision Box	To confirm	Configured options	Current technician status

Table 4.4: Field Specification – Technician Form

Calibration Schedule Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Schedule ID	Schedule_ID	Auto Number	To confirm	System generated	Unique schedule ID
Equipment	Equipment	Lookup	To confirm	Select equipment	Equipment reference
Laboratory	Laboratory	Lookup	To confirm	Select laboratory	Laboratory reference
Calibration Type	Calibration_Type	Drop Down	To confirm	Configured options	Type of calibration
Scheduled Date	Scheduled_Date	Date	To confirm	Date selection	Planned calibration date
Assigned Technician	Assigned_Technician	Lookup	To confirm	Select technician	Assigned technician
Priority	Priority	Decision Box	To confirm	Configured options	Schedule priority
Schedule Status	Schedule_Status	Drop Down	To confirm	Configured options	Current schedule status

Remarks	Remarks	Multi Line	To confirm	Optional	Additional information

Table 4.5: Field Specification – Calibration Schedule Form

Calibration Record Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Calibration Record ID	Calibration_ Record_ID	Auto Number	To confirm	System generated	Unique record ID
Equipment	Equipment	Lookup	To confirm	Select equipment	Equipment reference
Calibration Date	Calibration_ Date	Date	To confirm	Date selection	Calibration date
Technician	Technician	Lookup	To confirm	Select technician	Assigned technician
Calibration Method	Calibration_ Method	Drop Down	To confirm	Configured options	Method used
Calibration Resul	Calibration_ Result	Decision Box	To confirm	Configured options	Overall result
Overall Remarks	Overall_Rem arks	Multi Line	To confirm	Optional	Additional observations
Next Calibration Date	Next_Calibra tion_Date	Date	To confirm	Date selection	Next calibration date
Calibration Parameters	Calibration_P arameters	Subform	To confirm	Parameter entries	Calibration measurements

Table 4.6: Field Specification – Calibration Record Form

Calibration Parameters Subform

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Parameter Name	Parameter_Name	Single Line	To confirm	—	Parameter description
Standard Value	Standard_Value	Number	To confirm	Numeric value	Reference value
Actual Value	Actual_Value	Number	To confirm	Numeric value	Measured value
Tolerance	Tolerance	Number	To confirm	Numeric value	Permitted variation
Unit	Unit	Drop Down	To confirm	Configured options	Measurement unit
Result	Result	Decision Box	To confirm	Configured options	Parameter result
Remarks	Remarks	Single Line	To confirm	Optional	Additional notes

Table 4.7: Field Specification – Calibration Parameters Subform

Calibration Certificate Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Certificate ID	Certificate_ID	Auto Number	To confirm	System generated	Unique certificate ID
Certificate Number	Certificate_Number	Single Line	To confirm	—	Certificate reference number
Equipment	Equipment	Lookup	To confirm	Select equipment	Equipment reference
Calibration Record Form	Calibration_Record_Form	Lookup	To confirm	Select calibration record	Related calibration record
Issue Date	Issue_Date	Date	To confirm	Date selection	Certificate issue date
Valid From	Valid_From	Date	To confirm	Date selection	Start of validity

Valid Until	Valid_Until	Date	To confirm	Date selection	Expiry date

Certificate Status	Certificate_Status	Decision Box	To confirm	Configured options	Current certificate status
Certificate File	Certificate_File	File Upload	To confirm	File attachment	Uploaded certificate
Remarks	Remarks	Multi Line	To confirm	Optional	Additional details

Table 4.8: Field Specification – Calibration Certificate Form

Maintenance Request Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Maintenance ID	Maintenance_ID	Auto Number	To confirm	System generated	Unique maintenance ID
Equipment	Equipment	Lookup	To confirm	Select equipment	Equipment reference
Laboratory	Laboratory	Lookup	To confirm	Select laboratory	Laboratory reference
Reported By	Reported_By	Single Line	To confirm	—	Name of reporter
Request Date	Request_Date	Date	To confirm	Date selection	Request date
Problem Type	Problem_Type	Drop Down	To confirm	Configured options	Type of issue
Problem Description	Problem_Description	Multi Line	To confirm	—	Details of the problem
Priority	Priority	Decision Box	To confirm	Configured options	Maintenance priority
Assigned Technician	Assigned_Technician	Lookup	To confirm	Select technician	Assigned technician
Maintenance Status	Maintenance_Status	Drop Down	To confirm	Configured options	Current request status
Completion Date	Completion_Date	Date	To confirm	—	Maintenance completion date
Repair Cost	Repair_Cost	Currency	To confirm	Numeric value	Repair expenses

Repair Remarks	Repair_Remarks	Multi Line	To confirm	Optional	Repair details

Table 4.9: Field Specification – Maintenance Request Form

Equipment Inspection Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Inspection ID	Inspection_ID	Auto Number	To confirm	System generated	Unique inspection ID
Equipment Form	Equipment_Form	Lookup	To confirm	Select equipment	Equipment reference
Inspection Date	Inspection_Date	Date	To confirm	Date selection	Date of inspection
Inspector	Inspector	Single Line	To confirm	—	Name of inspector Name of inspector
Physical Condition	Physical_Condition	Decision Box	To confirm	Configured options	Physical condition
Functional Condition	Functional_Condition	Decision Box	To confirm	Configured options	Working condition
Safety Condition	Safety_Condition	Decision Box	To confirm	Configured options	Safety status
Inspection Result	Inspection_Result	Decision Box	To confirm	Configured options	Overall inspection result
Issues Found	Issues_Found	Multi Line	To confirm	Optional	Issues identified
Recommended Action	Recommended_Action	Drop Down	To confirm	Configured options	Suggested action
Inspection Remarks	Inspection_Remarks	Multi Line	To confirm	Optional	Additional observations

Table 4.10: Field Specification – Equipment Inspection Form

4.4 Process and Workflow Design

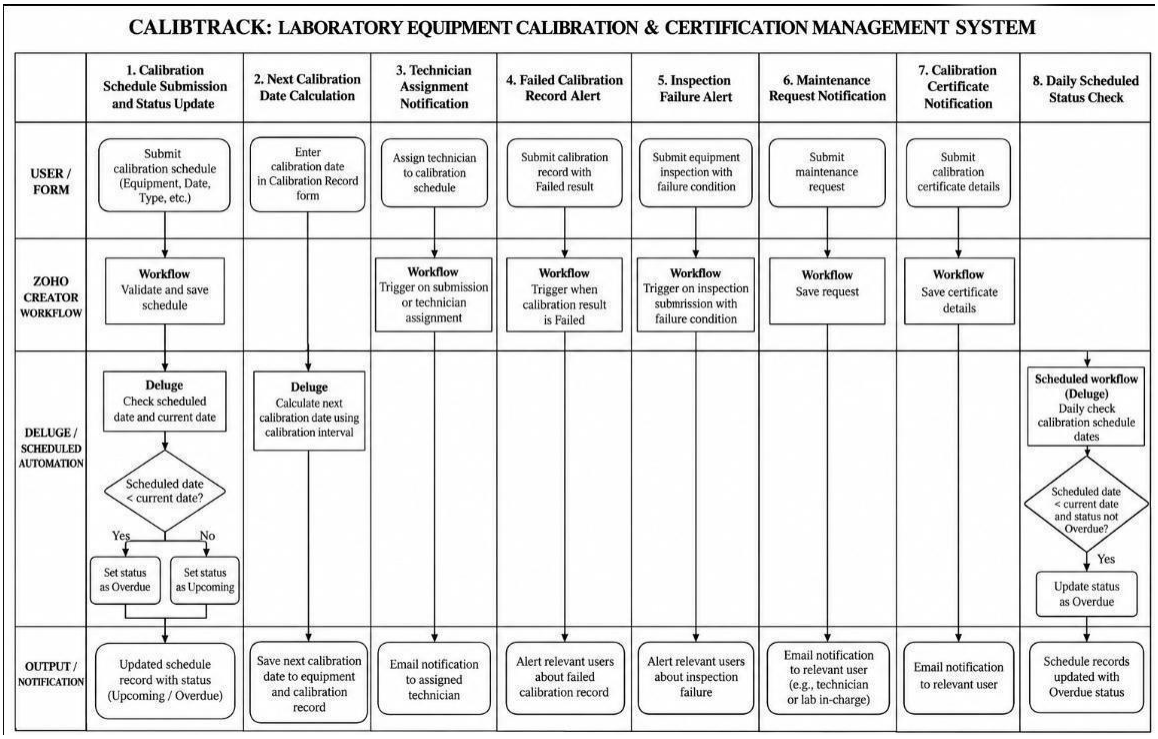


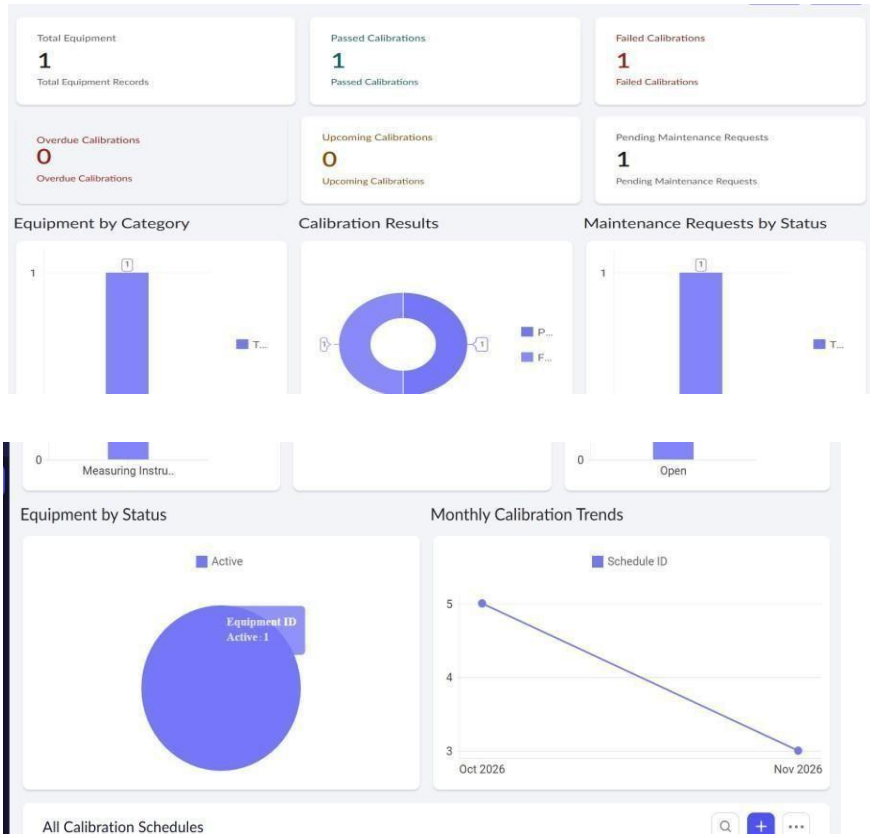
Figure 4.3: Configured Workflow and Process Flow

4.5 User Interface and Navigation Design





Figure 4.4(a): Navigation Structure of CALIBTRACK Application



All Calibration Schedules

Schedule ID	Equipment	Laboratory	Calibration Type	Scheduled Date	Assigned Technician
7	Digital Weighing Machine	Physics Laboratory	Routine Calibration	14-Oct-2026	Arun Kumar
6	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar
4	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar
3	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar
2	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar

Showing 5 of 5

All Equipment

Equipment ID	Equipment Name	Equipment Category	Laboratory	Manufacturer	Model Number
1	Digital Weighing Machine	Measuring Instrument	Physics Laboratory	ExampleTech	WM-100

All Maintenance Requests

Maintenance ID	Equipment	Laboratory	Reported By	Request Date	Problem Type	Pro
1	Digital Weighing Machine	Physics Laboratory	CALIBTRACK Admin	29-Oct-2026	Equipment Not Working	Tes imp

Showing 1 of 1



Figure 4.4(b): Calibration Dashboard of CALIBTRACK

4.6 Access Control Design

Complete the following matrix from the live permissions screen. Do not assume roles or permissions based only on menu visibility.

Component	CEO	Equipment Manager	Laboratory Technician	Calibration Supervisor	Read-only Auditor
Laboratory Form	A E V D	V	V	V	V
Equipment Form	A E V D	A E V	V	V	V
Technician Form	A E V D	V	V	V	V
Calibration Schedule Form	A E V D	A E V	A E V	V	V
Calibration Record Form	A E V D	V	A E V	V	V
Calibration Certificate Form	A E V D	V	A V	A E V	V

Maintenance Request Form	A E V D	A E V	A V	V	V
Equipment Inspection Form	A E V D	A E V	A E V	V	V
Reports	V	V	V	V	V
Calibration Dashboard	V	V	V	V	V

Table 4.11: Role–Permission Matrix

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

CALIBTRACK was developed using Zoho Creator, a low-code application development platform. The application was created in the Zoho Creator workspace using the form builder and other built-in features. The required forms, reports, workflows and Calibration Dashboard were configured to manage laboratory equipment and calibration activities. The application menu was organised into sections such as Laboratories, Equipment, Technicians, Calibration Schedules, Calibration Records, Calibration Certificates, Maintenance Requests and Equipment Inspections. A consistent application name and theme were used to make navigation simple and user-friendly.

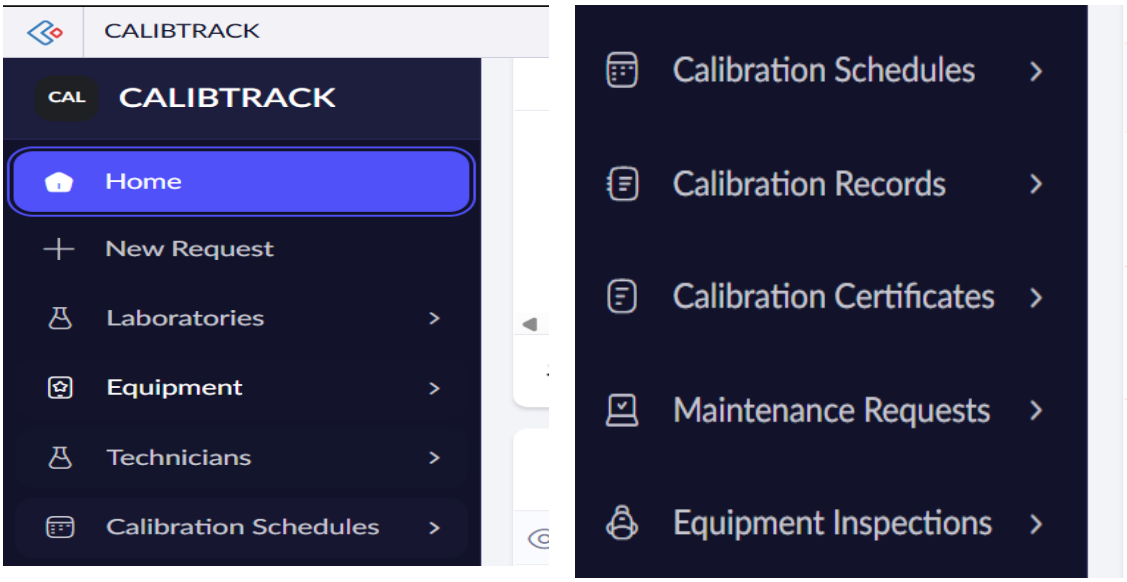


Figure 5.1: CALIBTRACK Application Home and Navigation

5.2 Forms and Data Capture

CALIBTRACK contains forms to collect and manage laboratory and equipment-related information. The application includes forms for Laboratory, Equipment, Technician, Calibration Schedule, Calibration Record, Calibration Certificate, Maintenance Request and Equipment Inspection. Each form contains fields to capture the required details, such as equipment information, dates, status, contact details and calibration results. Lookup fields connect related records, such as equipment with laboratories and technicians with calibration schedules. The Calibration Record form also includes a Calibration Parameters subform to enter individual measurement details. Mandatory fields, dropdowns and other field properties are configured to support data entry and maintain consistent records. These forms provide the main data entry facilities for the application.

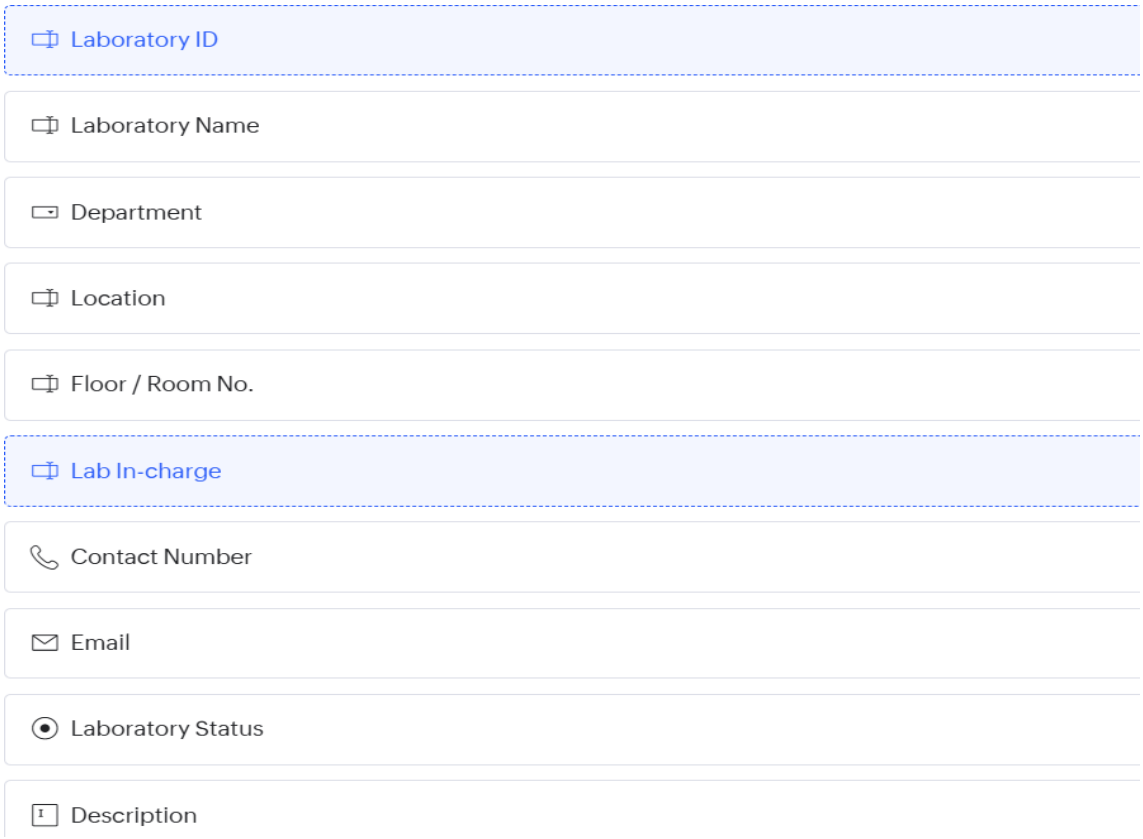


Figure 5.2: Laboratory Form in the Form Builder

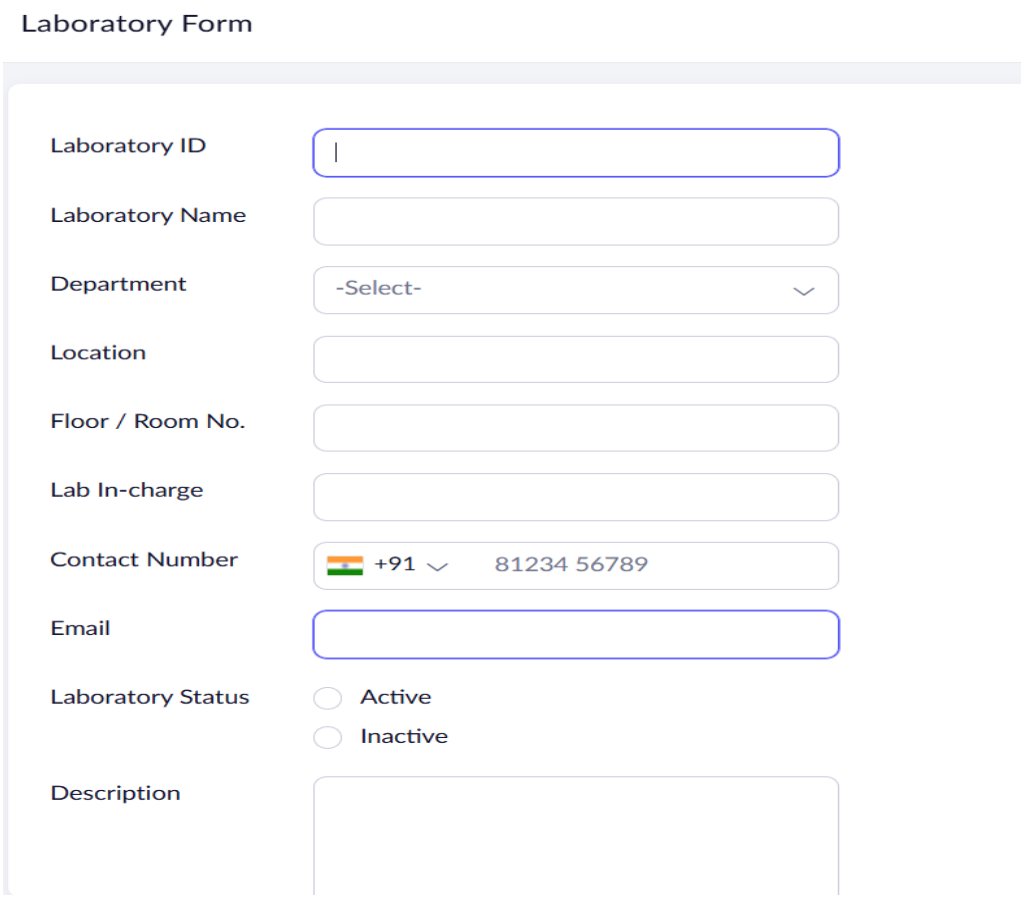


Figure 5.3: Laboratory Form in Live Mode

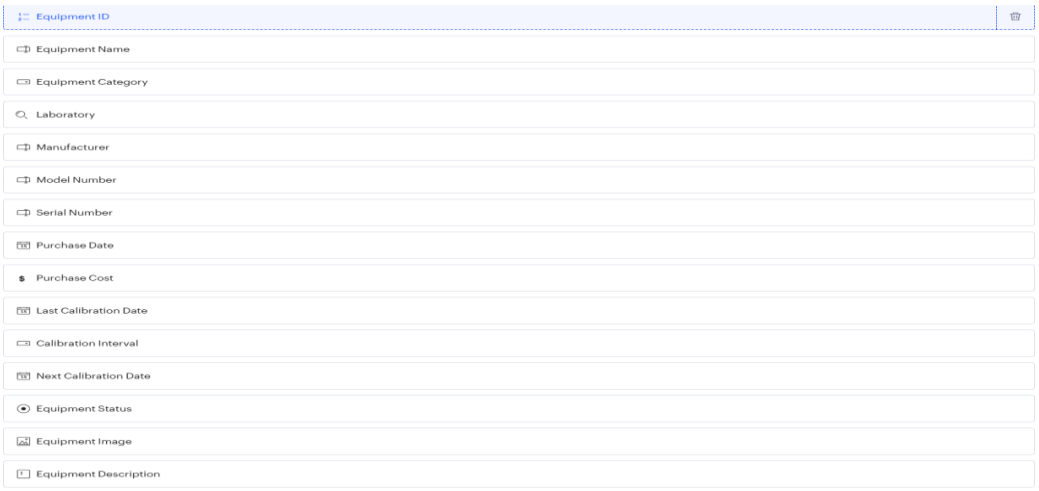


Figure 5.4: Equipment Form in the Form Builder

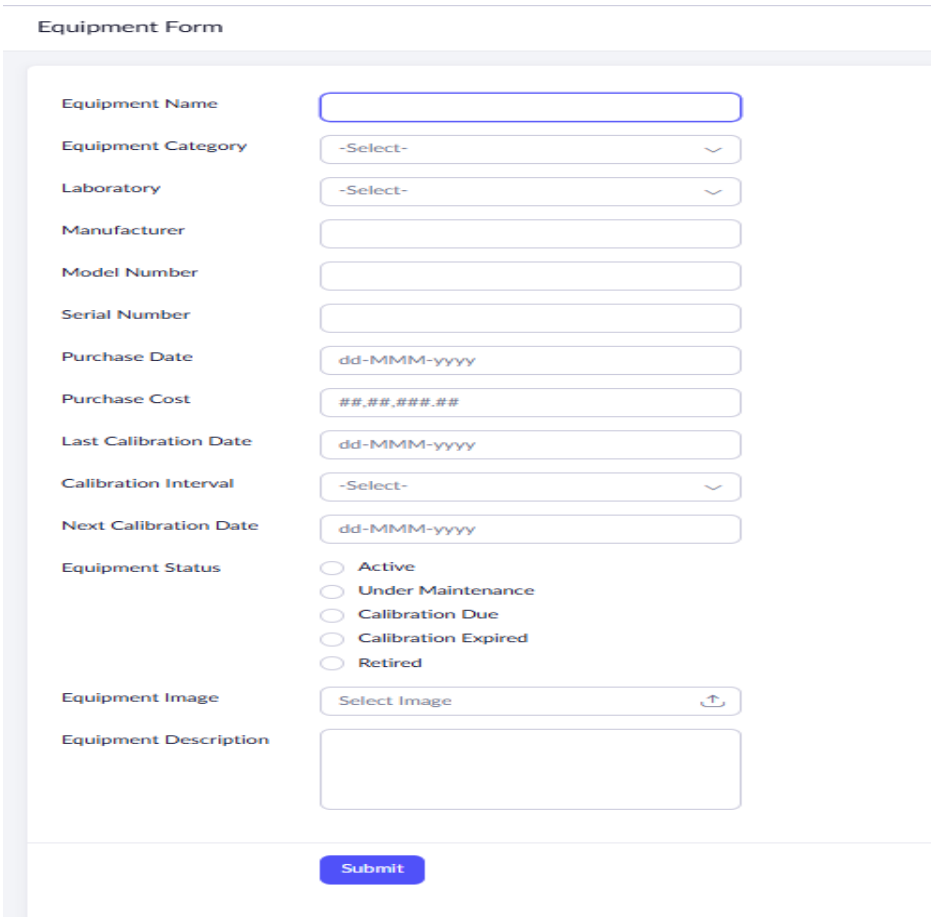


Figure 5.5: Equipment Form in Live Mode



Figure 5.6: Technician Form in the Form Builder

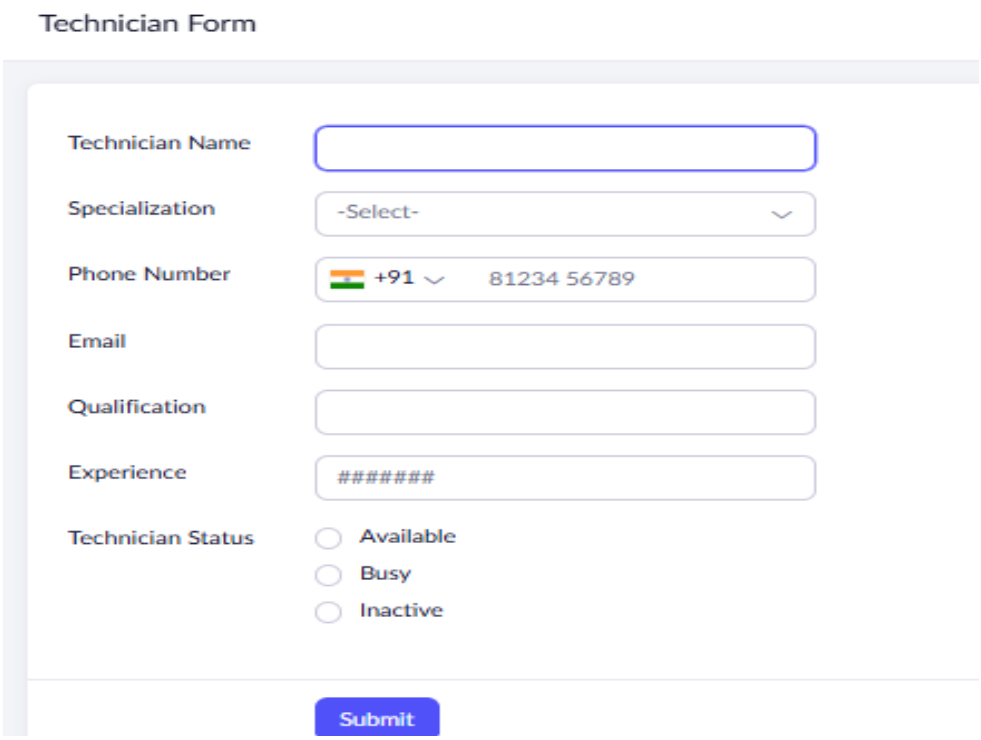


Figure 5.7: Technician Form in Live Mode

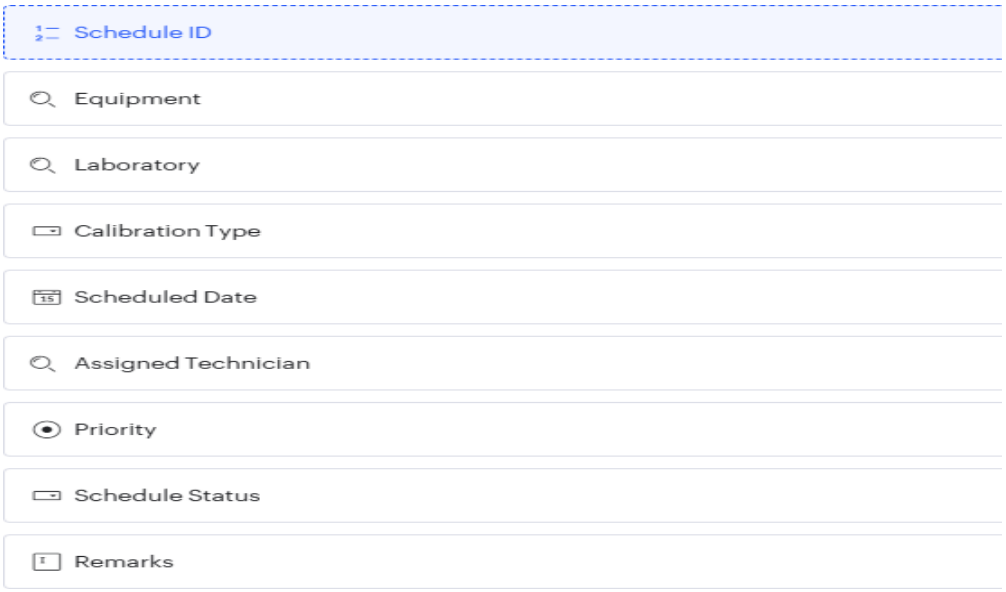


Figure 5.8: Calibration Schedule Form in the Form Builder

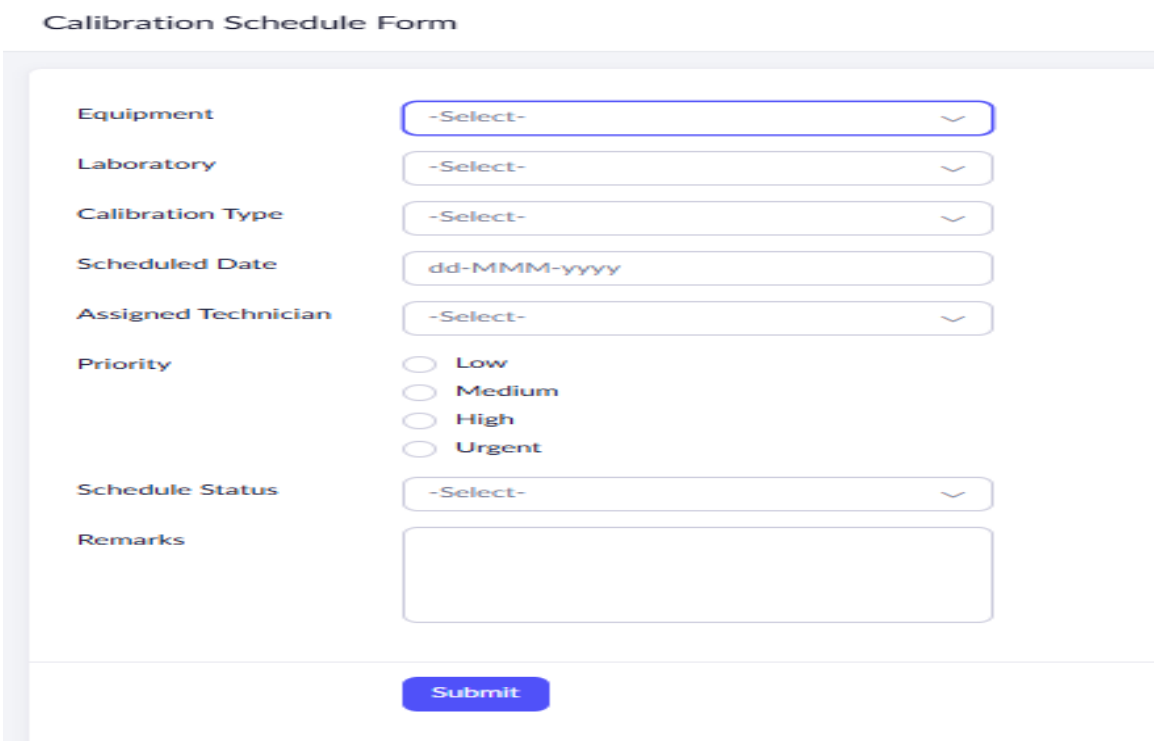


Figure 5.9: Calibration Schedule Form in Live Mode

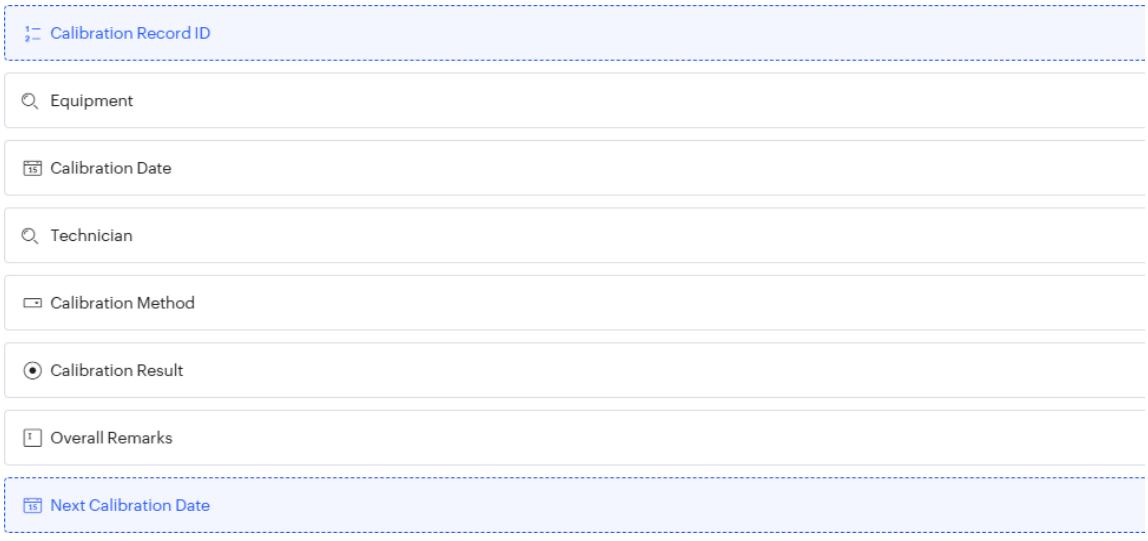


Figure 5.10: Calibration Record Form in the Form Builder

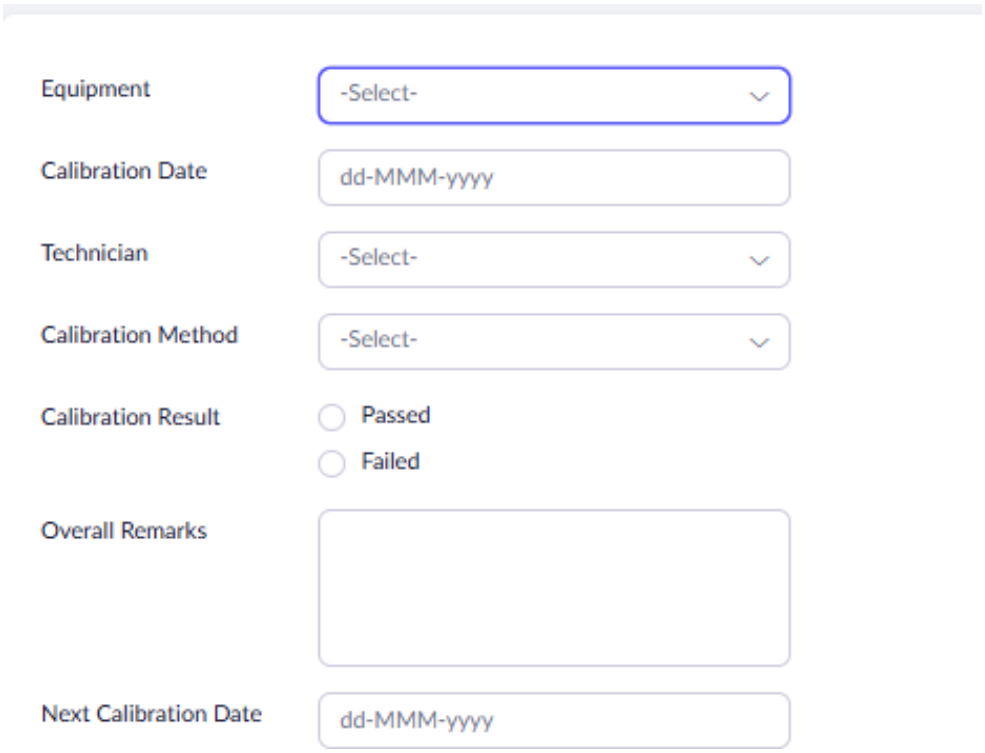


Figure 5.11: Calibration Record Form in Live Mode

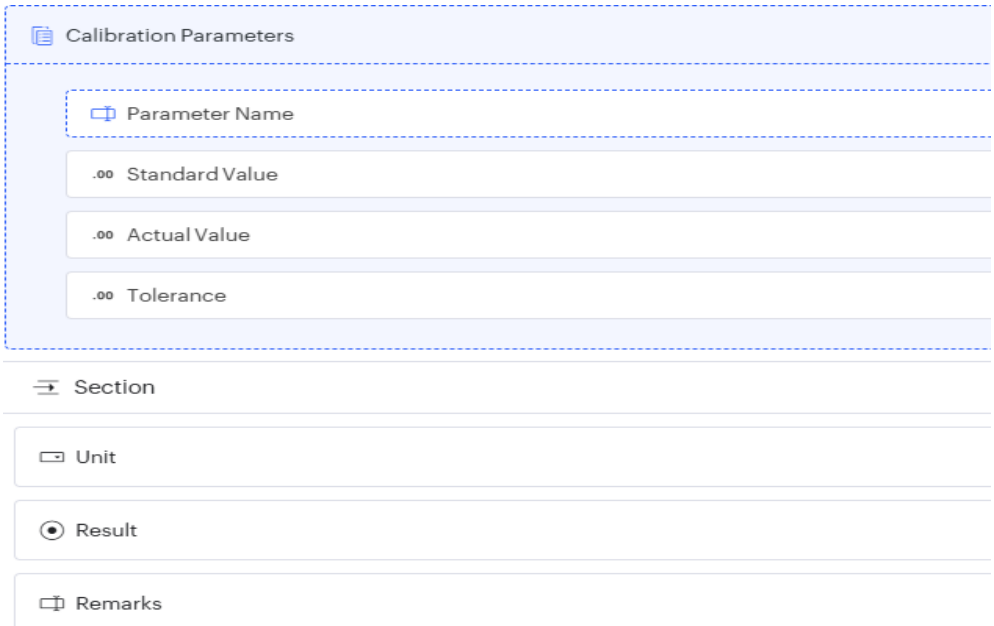


Figure 5.12: Calibration Parameters Subform in the Form Builder

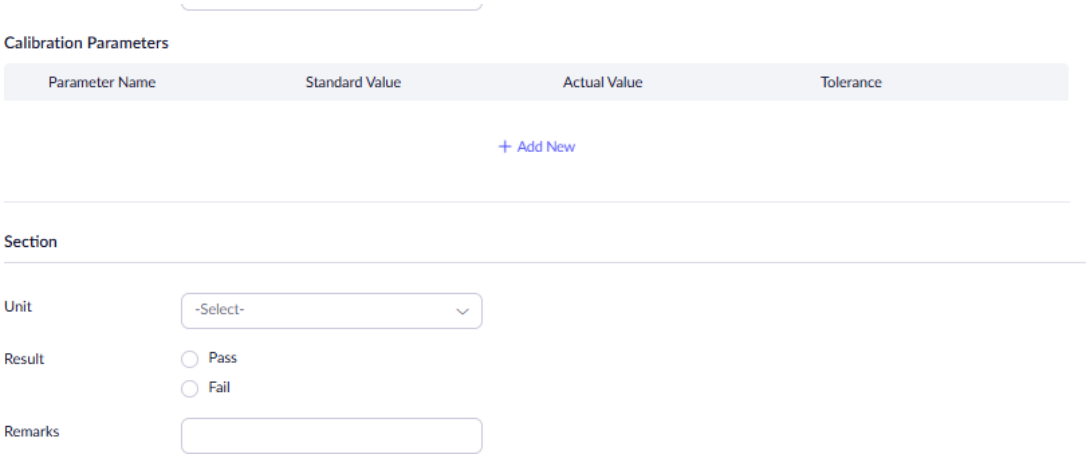


Figure 5.13: Calibration Parameters Subform in Live Mode

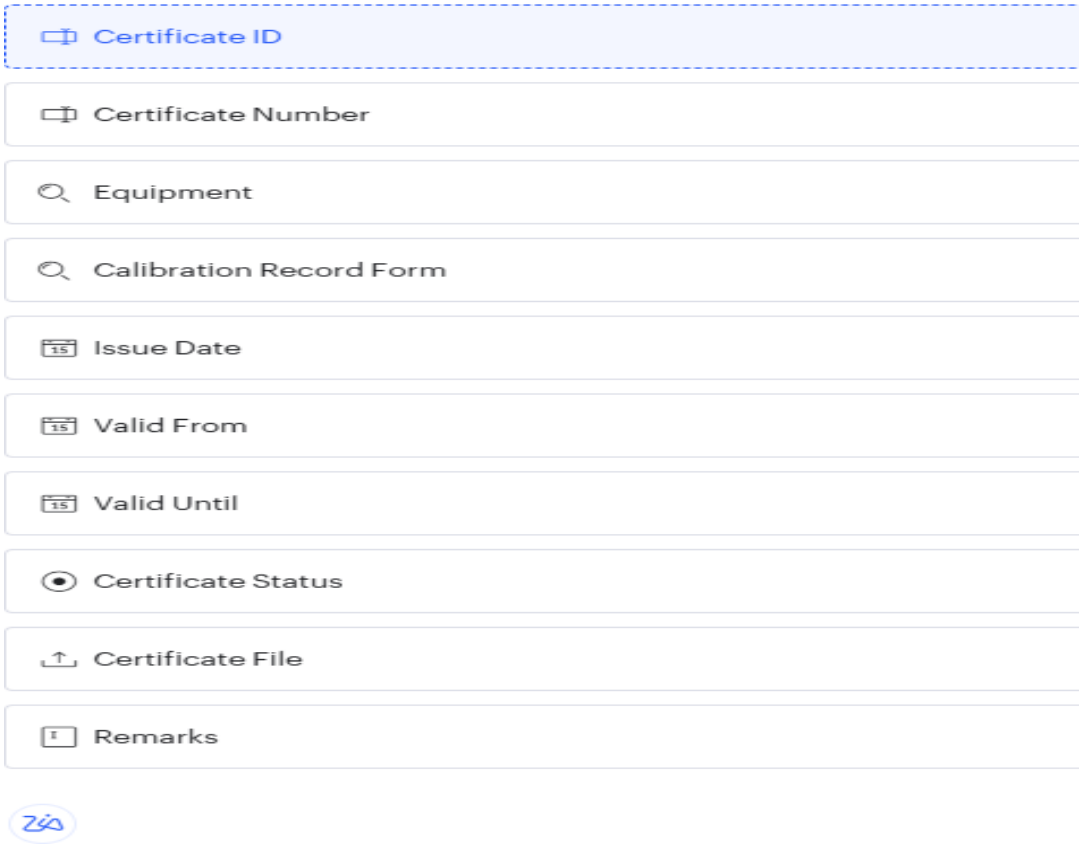


Figure 5.14: Calibration Certificate Form in the Form Builder

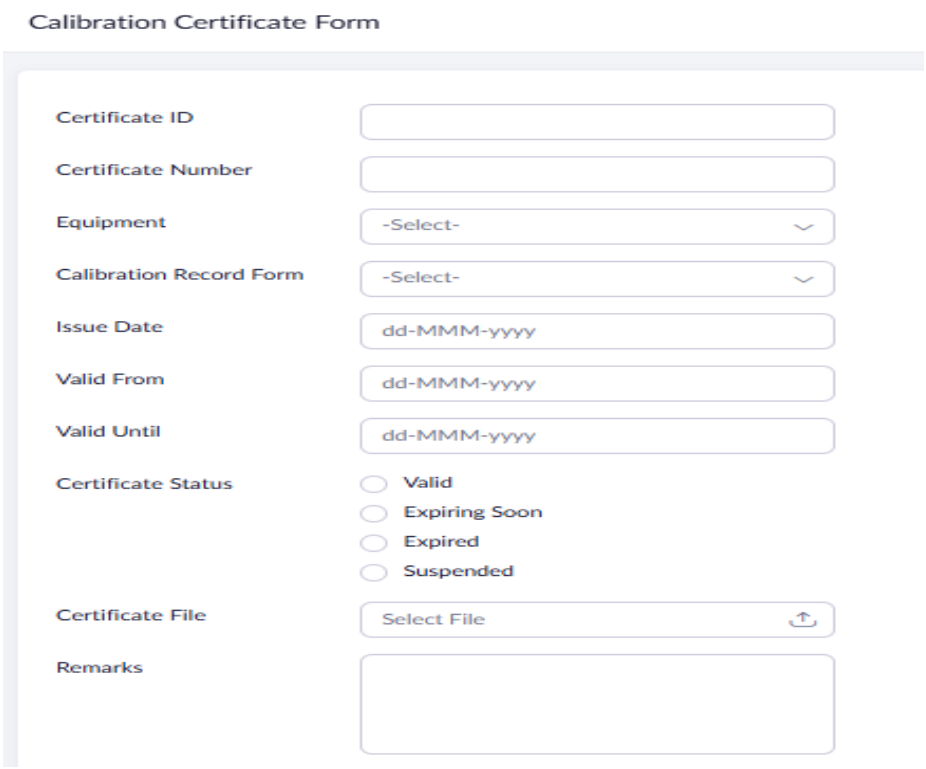


Figure 5.15: Calibration Certificate Form in Live Mode

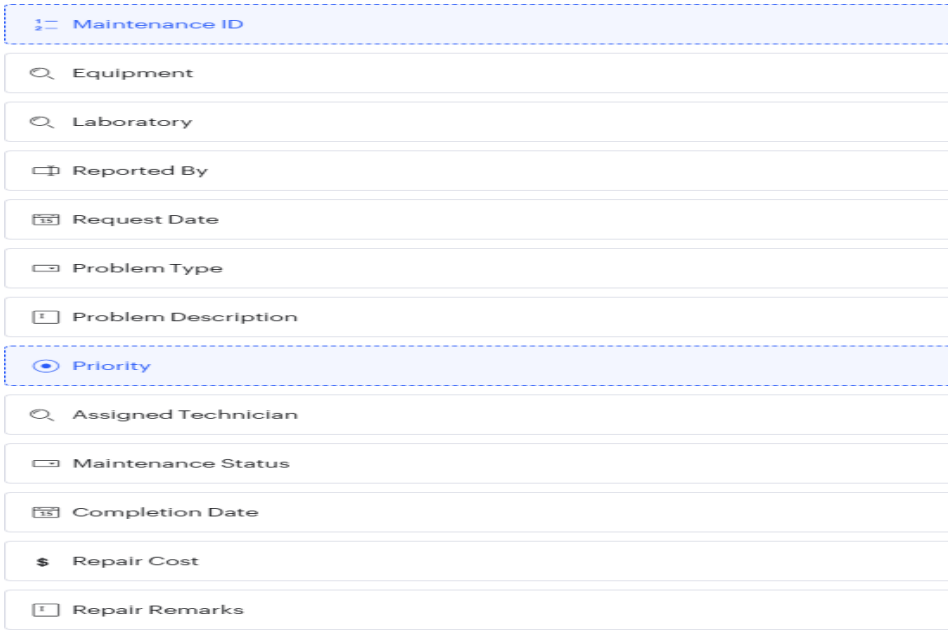


Figure 5.16: Maintenance Request Form in the Form Builder

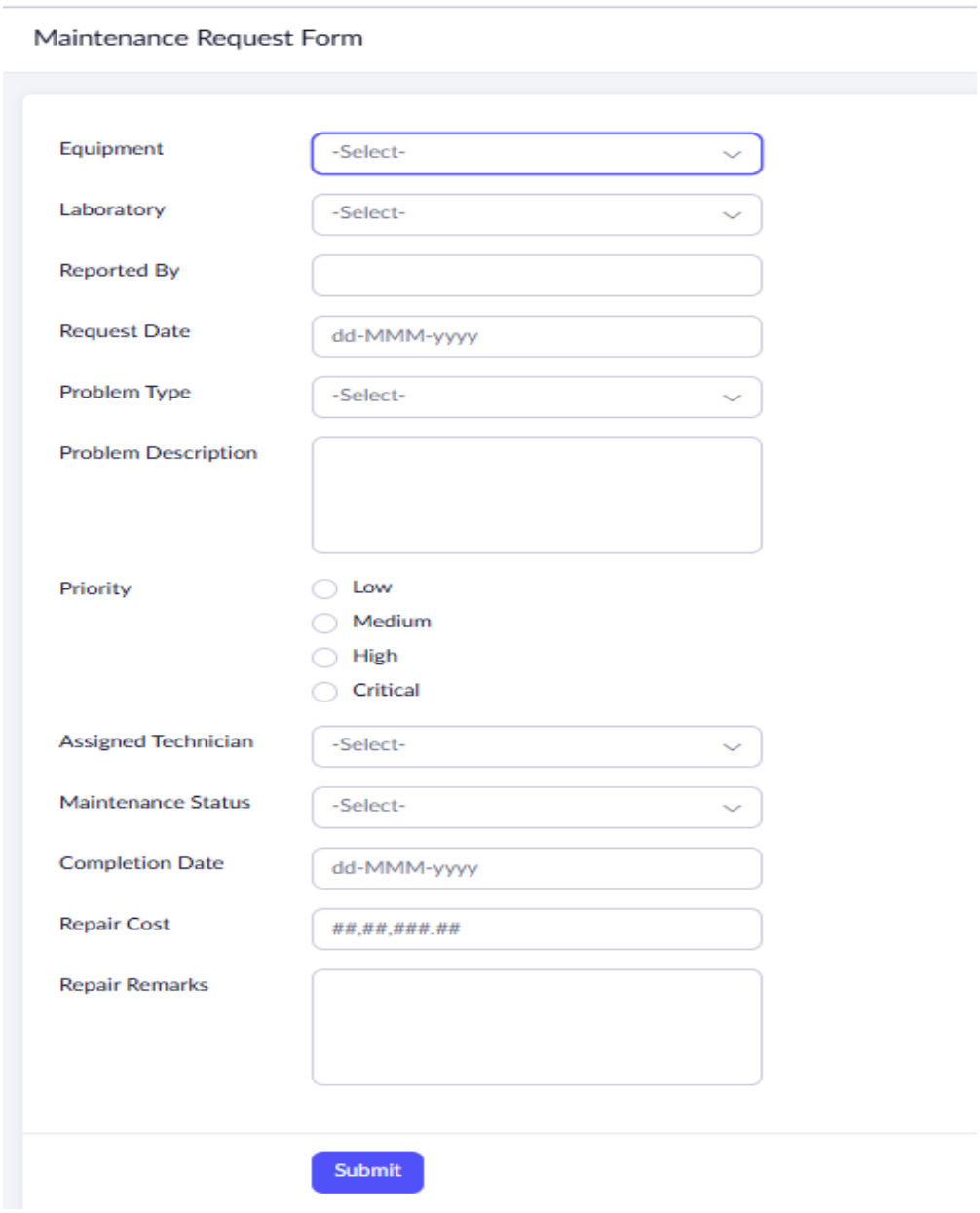


Figure 5.17: Maintenance Request Form in Live Mode



Figure 5.18: Equipment Inspection Form in the Form Builder

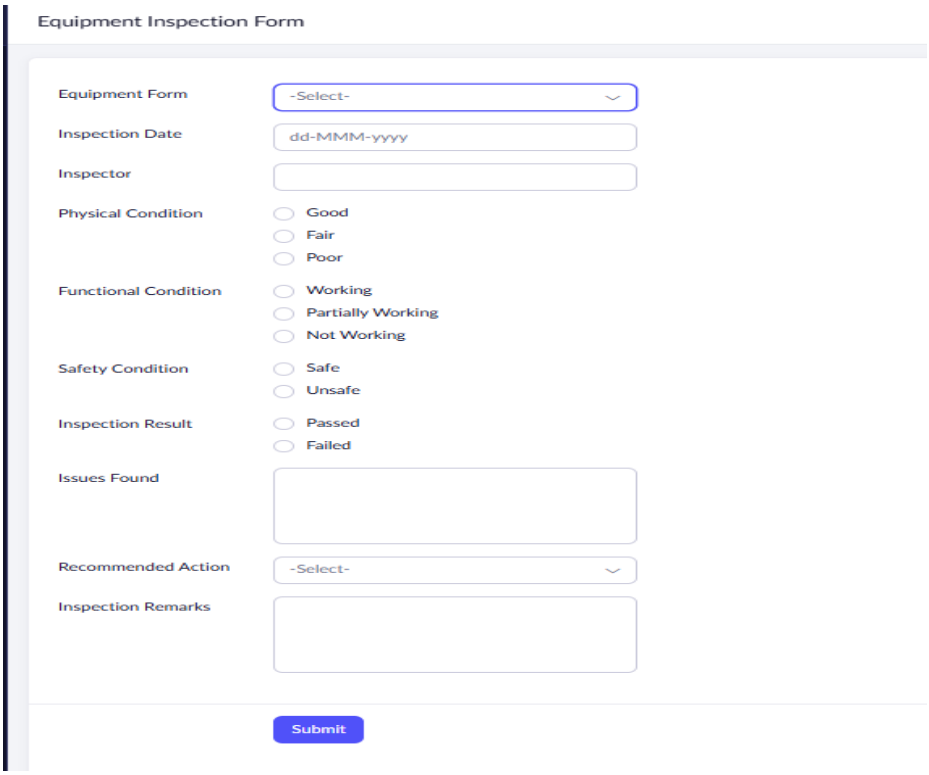


Figure 5.19: Equipment Inspection Form in Live Mode

5.3 Reports

Reports provide list-style access to stored records. The application navigation and dashboard confirm the presence of report areas for the core entities. The dashboard includes an All Equipment report with columns for Equipment ID, Equipment Name, Equipment Category, Laboratory, Manufacturer, Serial Number and Model Number. Confirm each report's exact name, type, filters, grouping, sorting, custom actions and access roles from the live app.

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Laboratories	List	Laboratory	Sort by Laboratory Name	View records	Authorised users
All Equipment	List	Equipment	Group by Equipment Category	View records	Authorised users
All Technicians	List	Technician	Sort by Technician Name	View records	Authorised users
All Calibration Schedules	List	Calibration Schedule	Sort by Scheduled Date	View records	Authorised users
All Calibration Records	List	Calibration Record	Sort by Calibration Date	View records	Authorised users
All Calibration Certificates	List	Calibration Certificate	Sort by Issue Date	View records	Authorised users
All Maintenance Requests	List	Maintenance Request	Group by Maintenance Status	View records	Authorised users
All Equipment Inspections	List	Equipment Inspection	Sort by Inspection Date	View records	Authorised users
Calibration History Report	List	Calibration Record	Sort by Calibration Date	View history	Authorised users
Calibration Schedule Report	List	Calibration Schedule	Group by Schedule Status	View schedules	Authorised users

Calibration Status Report	List	Calibration Record	Group by Calibration Result	View status	Authorised users
Certificate Expiry Report	List	Calibration Certificate	Sort by Valid Until	View expiry details	Authorised users
Equipment Report	List	Equipment	Group by Equipment Status	View equipment details	Authorised users
Maintenance Report	List	Maintenance Request	Group by Maintenance Status	View maintenance details	Authorised users
Technician Workload Report	List	Calibration Schedule	Group by Assigned Technician	View assigned schedules	Authorised users

Table 5.1: Reports Implemented

Equipment ID	1
Equipment Name	Digital Weighing Machine
Equipment Category	Measuring Instrument
Laboratory	Physics Laboratory
Manufacturer	ExampleTech
Model Number	WM-100
Serial Number	SN1001
Purchase Date	15-Jan-2025
Purchase Cost	₹ 15,000.00
Last Calibration Date	15-Oct-2026
Calibration Interval	6 Months
Next Calibration Date	15-Apr-2027
Equipment Status	Active
Equipment Image	
Equipment Description	Digital weighing machine used for laboratory measurements and calibration testing.

Figure 5.20: Equipment Report

Schedule ID	Equipment	Laboratory	Calibration Type	Scheduled Date	Assigned Technician	Priority
8	Digital Weighing Machine	Physics Laboratory	Routine Calibration	03-Nov-2026	Arun Kumar	
7	Digital Weighing Machine	Physics Laboratory	Routine Calibration	14-Oct-2026	Arun Kumar	Medium
6	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar	
5	Digital Weighing Machine	Physics Laboratory	Routine Calibration	18-Nov-2026	Arun Kumar	Medium
4	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar	Medium
3	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar	Low
2	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar	Low
1	Digital Weighing Machine	Physics Laboratory	Routine Calibration	05-Nov-2026	Arun Kumar	High

Figure 5.21: Calibration Schedules Report

Calibration History Report

Calibration Record ID	Equipment	Calibration Date	Technician	Calibration Method	Calibration Result	Overall Remarks
2	Digital Weighing Machine	29-Oct-2026	DEVA	Routine Calibration	Passed	
1	Digital Weighing Machine	29-Oct-2026	Arun Kumar	Routine Calibration	Failed	Test failure alert workflow.

Figure 5.22: Calibration History Report

Calibration Status Report

Calibration Record ID	Equipment	Calibration Date	Technician	Calibration Method	Calibration Result	Overall Remarks
2	Digital Weighing Machine	29-Oct-2026	DEVA	Routine Calibration	Passed	
1	Digital Weighing Machine	29-Oct-2026	Arun Kumar	Routine Calibration	Failed	Test failure alert workflow.

Figure 5.23: Calibration Status Report

Certificate Expiry Report

Certificate ID	Certificate Number	Equipment	Calibration Record Form	Issue Date	Valid From	Valid Until
2	2	Digital Weighing Machine	1	30-Oct-2026	30-Oct-2026	30-Oct-2026
CERT001	CAL-2026-001	Digital Weighing Machine	1	29-Oct-2026	29-Oct-2026	29-Oct-2027

Figure 5.24: Certificate Expiry Report

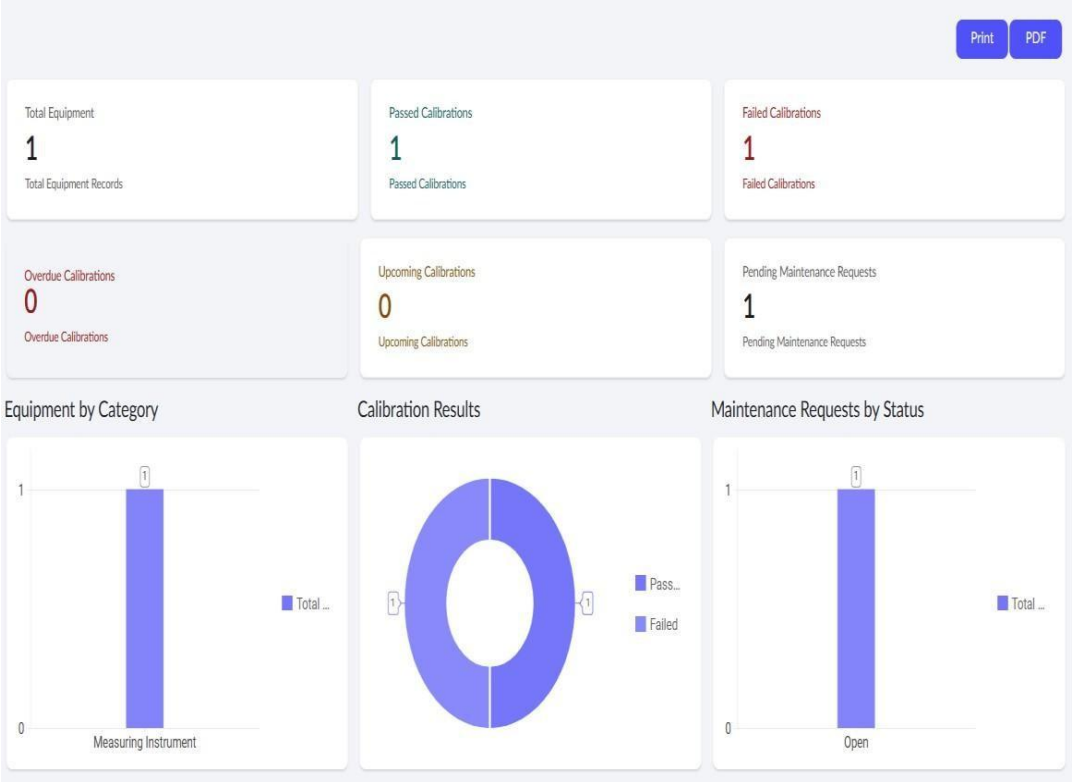
Maintenance Report

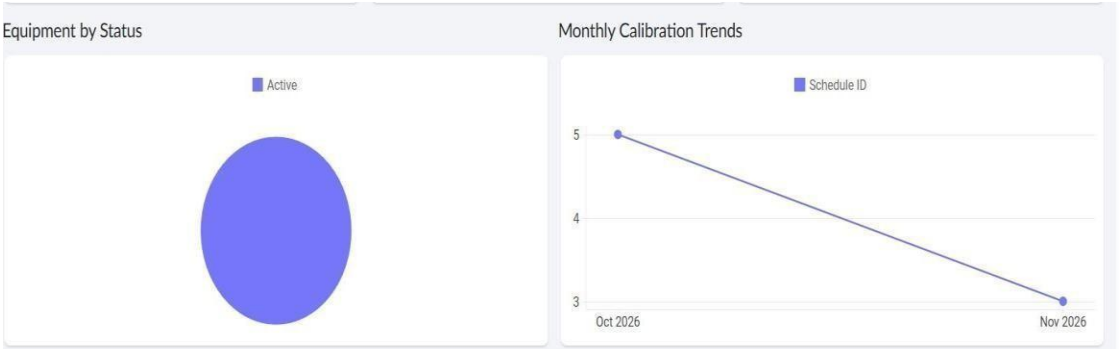
Maintenance ID	Equipment	Laboratory	Reported By	Request Date	Problem Type	Problem Description
1	Digital Weighing Machine	Physics Laboratory	CALIBTRACK Admin	29-Oct-2026	Equipment Not Working	Testing maintenance request notification after applicatior

Figure 5.25: Maintenance Report

5.4 Pages and Dashboards

CALIBTRACK contains a page named Calibration Dashboard that provides a quick overview of laboratory equipment and calibration activities. It displays KPI cards for total equipment, passed calibrations, failed calibrations, overdue calibrations, upcoming calibrations and pending maintenance requests. The dashboard also includes charts for equipment status, calibration results, maintenance requests and monthly calibration trends. Embedded reports help users view equipment, calibration schedules and maintenance details. This dashboard helps users monitor activities and identify pending tasks from one place.





All Calibration Schedules

Schedule ID	Equipment	Laboratory	Calibration Type	Scheduled Date	Assigned Technician	Priority
7	Digital Weighing Machine	Physics Laboratory	Routine Calibration	14-Oct-2026	Arun Kumar	Medium
6	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar	
4	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar	Medium
3	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar	Low
2	Digital Weighing Machine	Physics Laboratory	Routine Calibration	29-Oct-2026	Arun Kumar	Low

Showing 5 of 5

All Equipment

Equipment ID	Equipment Name	Equipment Category	Laboratory	Manufacturer	Model Number	Serial Number
1	Digital Weighing Machine	Measuring Instrument	Physics Laboratory	ExampleTech	WM-100	SN1001

Showing 1 of 1

All Calibration Schedules

Schedule ID	Equipment	Laboratory	Calibration Type	Scheduled Date	Assigned Technician	Priority
8	Digital Weighing Machine	Physics Laboratory	Routine Calibration	03-Nov-2026	Arun Kumar	
5	Digital Weighing Machine	Physics Laboratory	Routine Calibration	18-Nov-2026	Arun Kumar	Medium
1	Digital Weighing Machine	Physics Laboratory	Routine Calibration	05-Nov-2026	Arun Kumar	High

Showing 3 of 3

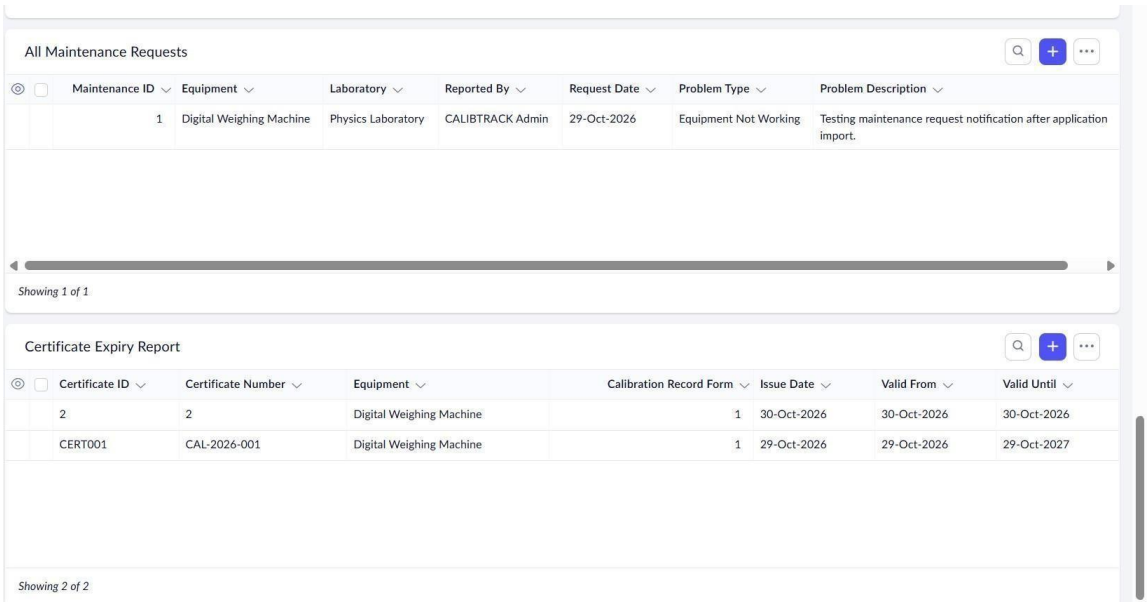


Figure 5.26: CALIBTRACK Calibration Dashboard

5.5 Deluge Scripting

Deluge is used for selected workflow logic. The technician-assignment notification path was configured to retrieve the assigned technician record and send an email to the technician's email address; the user reported that the email arrived during testing. The exact script must be copied from the live workflow into Appendix C. A failed-calibration alert workflow is also reported as tested. This report intentionally does not invent complete source code or field link names.

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Calibration Record Form – On User Input of Calibration Date	Automatically calculates the next calibration date by adding six months to the entered calibration date.	If condition, null check, addMonth(), field assignment
DS-02	Calibration Schedule Form – On Successful Form Submission	Automatically changes the schedule status to Overdue when the scheduled date is earlier than the current date.	If condition, date comparison, current date, field assignment

DS-03	Calibration Schedule Form – On Successful Form Submission	Retrieves the assigned technician's email address and sends an email notification about the assignment.	Record lookup, count(), if condition, variable assignment, sendmail
DS-04	Calibration Certificate Form – On Successful Form Submission	Checks the certificate validity date and sends an email alert when the certificate is approaching expiry within seven days.	If condition, date comparison, addDay(), sendmail

Table 5.2: Deluge Scripts Implemented

fx Deluge script

```
1  if(input.Calibration_Date != null)
2  {
3      input.Next_Calibration_Date = input.Calibration_Date.addMonth(6);
4  }
5
```

Code 5.1: Next Calibration Date Calculation

fx Deluge script

```
1  if(input.Scheduled_Date < zoho.currentdate)
2  {
3      input.Schedule_Status = "Overdue";
4  }
5
```

Code 5.2: Calibration Schedule Status Check

```
technicianRecord = Technician_Form[ID == input.Assigned_Technician];
if(technicianRecord.count() > 0)
{
    technicianEmail = technicianRecord.Email;
    if(technicianEmail != null && technicianEmail != "")
    {
        sendmail
        [
            from :zoho.adminuserid
            to :technicianEmail
            subject :"New Calibration Task Assigned - CALIBTRACK"
            message :"Dear Technician,<br><br>A new calibration task has been assigned to you. Please log in to CALIBTRACK and check your calibration schedule.<br><br>Thank you,<br>CALIBTRACK System"
        ]
    }
}
```

Code 5.3: Technician Assignment Notification

```
if(input.Valid_Until <= zoho.currentdate.addDay(7))
{
    sendmail
    [
        from :zoho.loginuserid
        to :"dharshini.g2.2024.cse@rajalakshmi.edu.in"
        subject :"CALIBTRACK - Certificate Expiry Alert"
        message :"Dear Admin,\n\nA calibration certificate is expiring soon.\n\nCertificate ID: " + input.Certificate_ID + "\nCertificate Number: " + input.Certificate_Number + "\nValid Until: " + input.Valid_Until + "\n\nPle
arrange certificate renewal.\n\nRegards,\nCALIBTRACK System"
    ]
}
```

Code 5.4: Certificate Expiry Notification

5.6 Workflows and Automation

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Calculate Next Calibration Date	Form Workflow	On User Input – Calibration Date	Calibration Date is not empty	Calculate and update the next calibration date by adding 6 months	DS-01
Calibration Record Submission Workflow	Form Workflow	On Successful Form Submission	No additional condition	Execute configured submission actions and notification	—

Failed Calibration Alert	Form Workflow	On Successful Form Submission	Calibration Result = Failed	Send email alert	—
Equipment Status on Failure	Form Workflow	On Successful Form Submission	Calibration failure condition	Update equipment status	—
Calibration Schedule Status Check	Form Workflow	On Successful Form Submission	Scheduled Date is earlier than current date	Update Schedule Status to Overdue	DS-02
Technician Assignment Notification	Form Workflow	On Successful Form Submission	No additional condition	Send email notification to assigned technician	DS-03
Certificate Submission Notification	Form Workflow	On Successful Form Submission	Certificate Valid Until date is within the next 7 days	Send certificate expiry notification	DS-04
Maintenance Request Notification	Form Workflow	On Successful Form Submission	No additional condition	Send maintenance request notification	—
Technician Registration Notification	Form Workflow	On Successful Form Submission	No additional condition	Send technician registration notification	—

Equipment Inspection Failure Alert	Form Workflow	On Successful Form Submission	Inspection failure condition	Send email alert	—
Daily Calibration Status Check	Form Workflow	Daily scheduled execution	Calibration schedule criteria	Check calibration schedules and perform configured actions	—

Table 5.3: Workflows Implemented

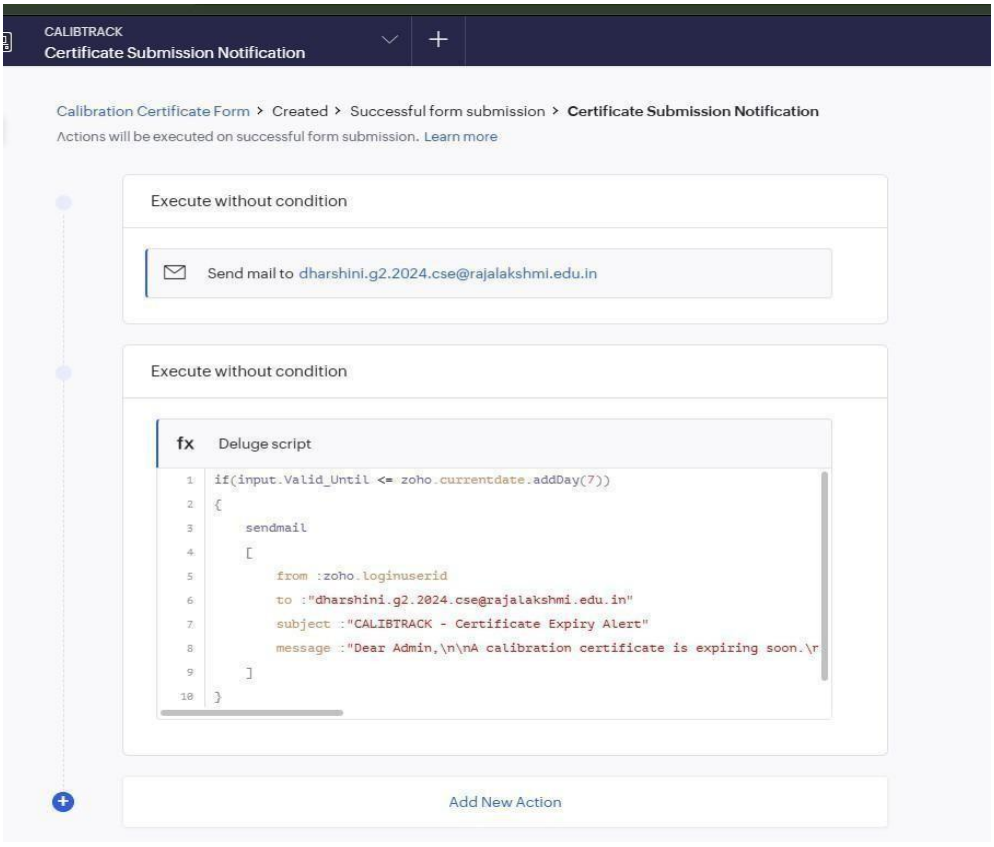


Figure 5.27: Certificate Submission Notification Workflow

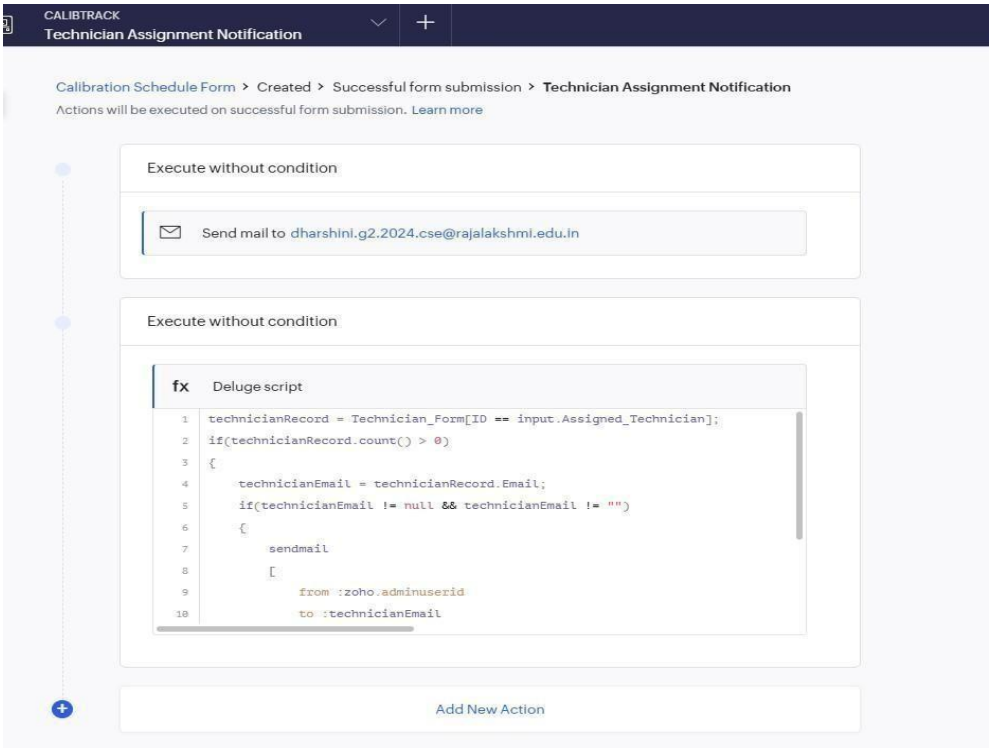


Figure 5.28: Technician Assignment Notification Workflow

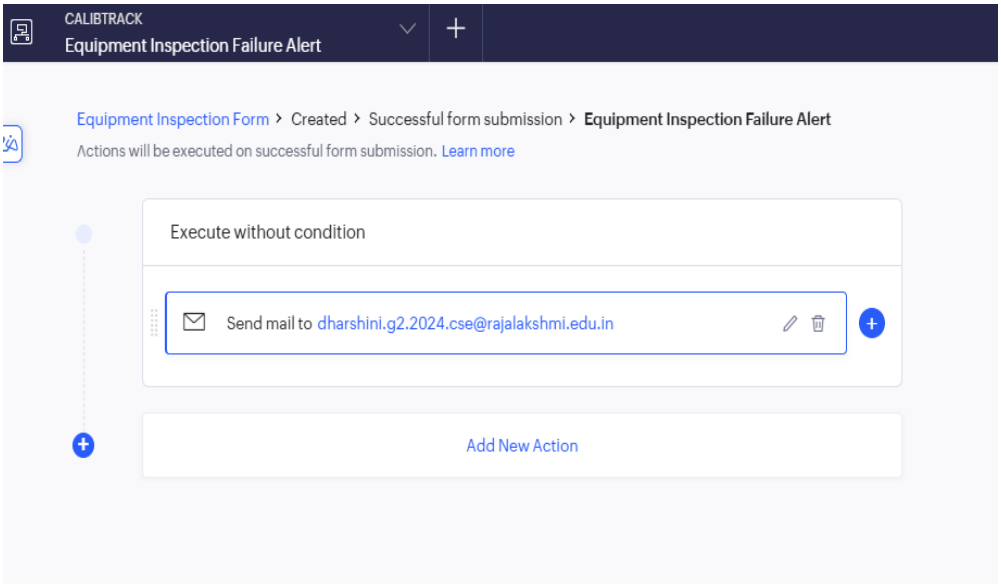


Figure 5.29: Equipment Inspection Failure Alert Workflow

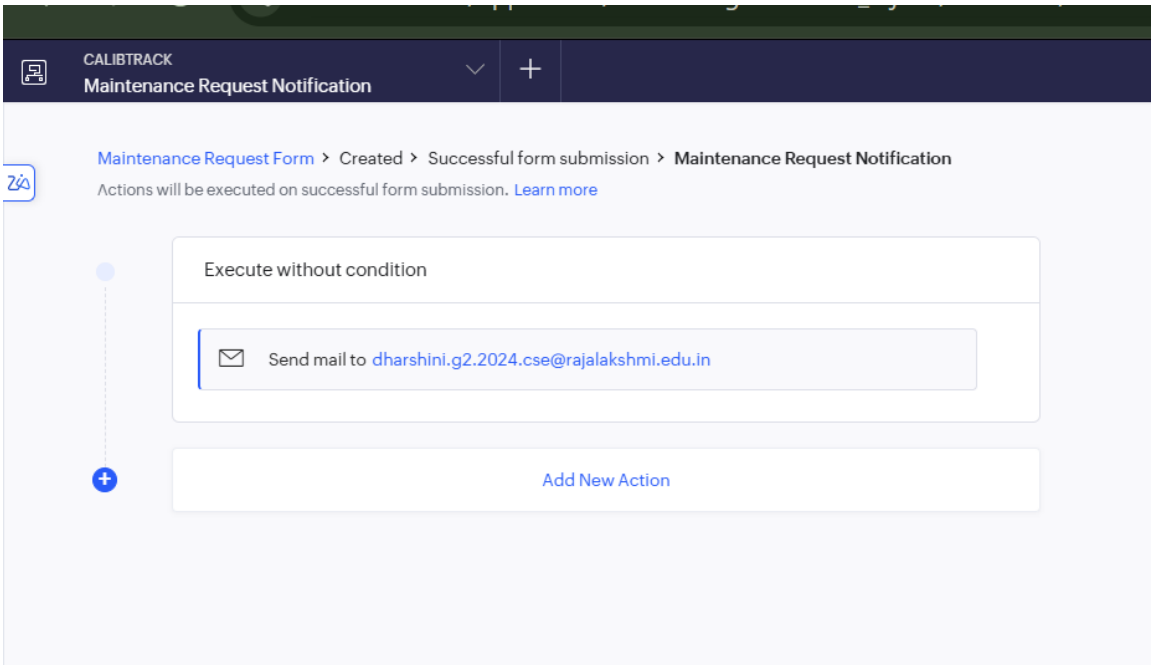


Figure 5.30: Maintenance Request Notification Workflow

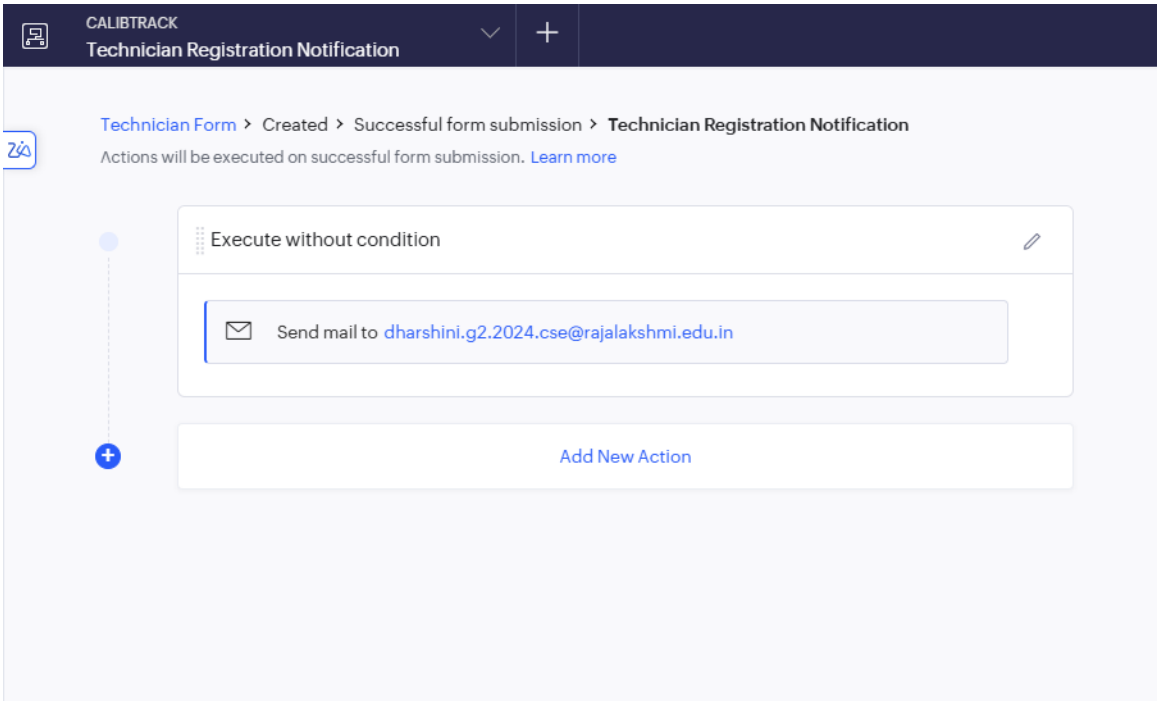


Figure 5.31: Technician Registration Notification Workflow

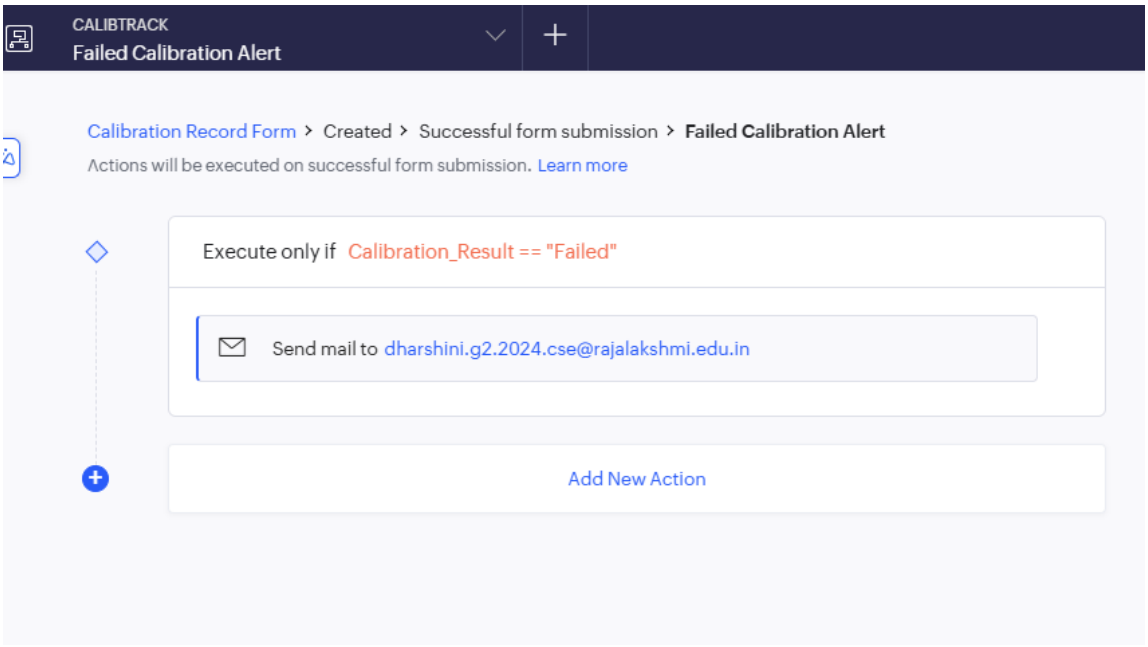


Figure 5.32: Failed Calibration Alert Workflow

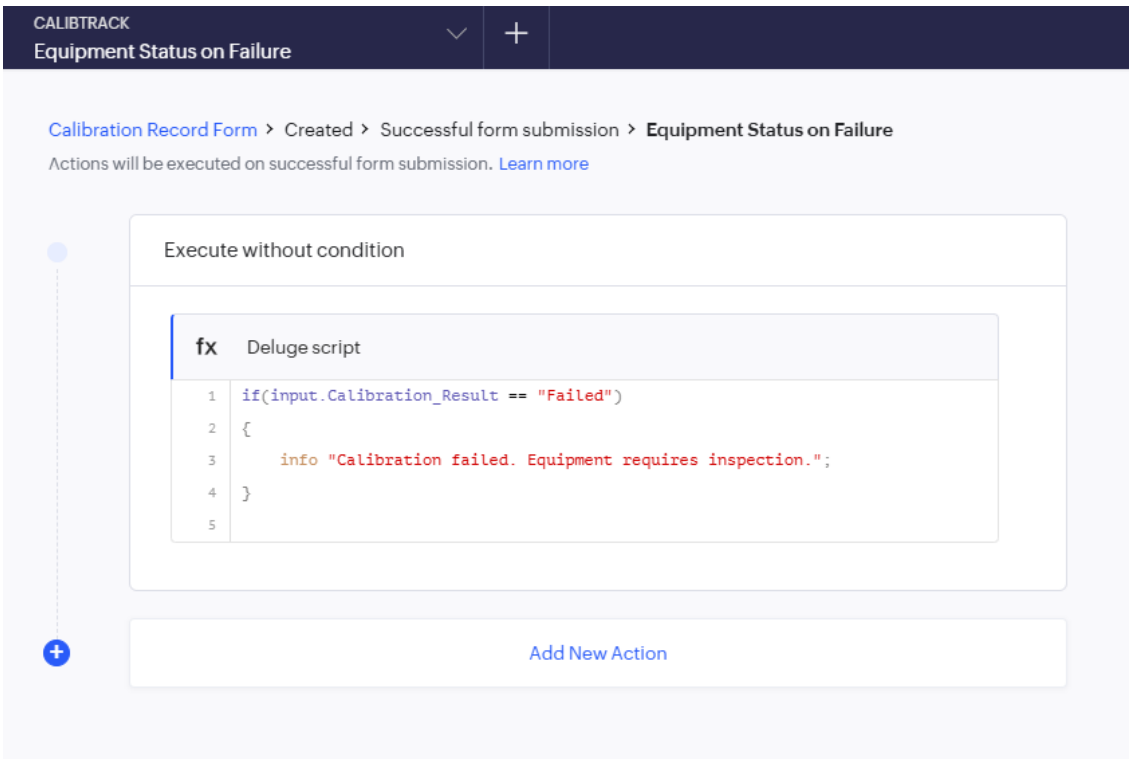


Figure 5.33: Equipment Status on Failure Workflow

5.7 Integrations

CALIBTRACK uses the built-in email notification functionality provided by Zoho Creator. Email notifications are configured for selected events, such as technician assignment, certificate expiry and failed calibration. Deluge scripts are also used in selected workflows to support these notifications. No external REST APIs, CRM connections or third-party integrations have been implemented in this project.

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Email Notifications	Zoho Creator	Built-in Send Mail action	Zoho Creator account	Sends notifications for selected calibration and maintenance events

Table 5.4: Integrations Implemented

5.8 User Management and Role-Based Access

CALIBTRACK includes user roles and permission profiles to organise access to the application. The configured roles include CEO, Laboratory Technician, Equipment Manager, Calibration Supervisor and Read-only Auditor. Permission profiles are created to define access levels for different users. These settings help organise user access to forms, reports and application features.

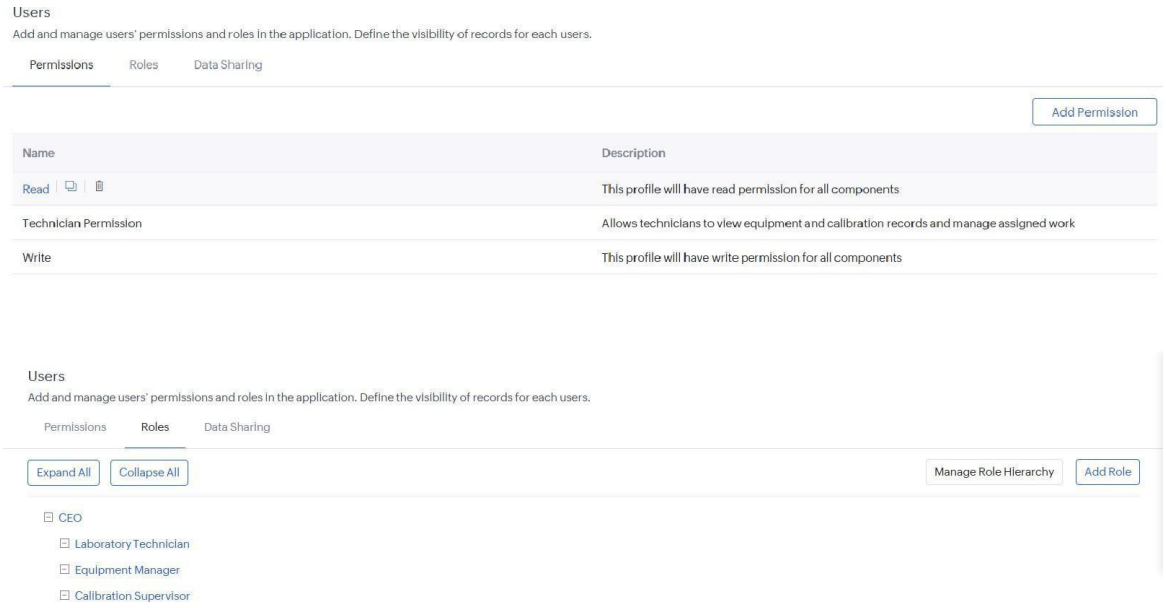


Figure 5.34: User Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

Zoho Creator provides AI-assisted features that support application development. In CALIBTRACK, Zia was used to assist with improving the appearance and arrangement of the dashboard KPI cards. The dashboard also includes charts to display equipment categories, calibration results, maintenance request status, equipment status and monthly calibration trends. These charts are based on the application records. The dashboard layout was reviewed after the changes. No separate AI prediction, OCR or external Zoho Analytics integration was implemented.

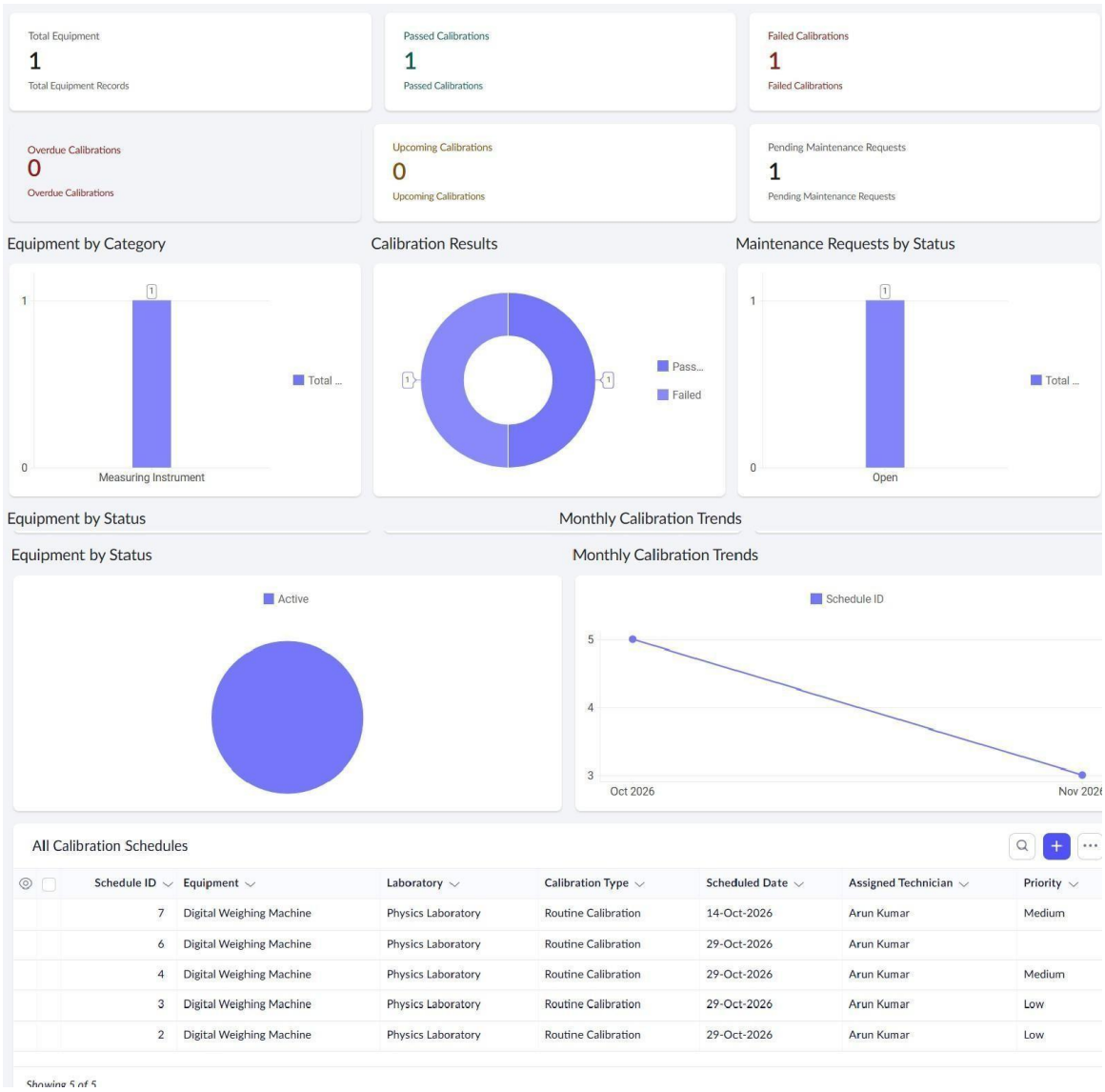


Figure 5.35: CALIBTRACK Dashboard – AI-Assisted and Analytics Feature

5.10 Pro-Code Extension

CALIBTRACK was developed mainly using Zoho Creator's low-code features. Deluge scripting was used in selected workflows to automate date calculations, update calibration schedule status and support email notifications. Advanced extensions such as Zoho Catalyst, JavaScript widgets and external REST API integrations were not implemented.

These features could be considered in future development to extend the application's functionality

5.11 Mobile and Portal Access

CALIBTRACK is developed using Zoho Creator and can be accessed through a supported web browser. Mobile access through the Zoho Creator mobile application and customer portal access were not separately configured or tested as part of this project.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out to check the forms, workflows, Deluge scripts and dashboard features of the CALIBTRACK application. Each form was reviewed to ensure that the fields, data types and lookup relationships were configured properly. The form submission process was checked to confirm that records could be entered and stored.

Selected Deluge scripts were tested to verify their functions. The next calibration date calculation was checked, and the calibration schedule status script was tested to confirm that past scheduled dates were marked as overdue. Workflow configurations were also reviewed to check their triggers and actions.

The dashboard was checked to ensure that KPI cards and charts displayed information from the application records. Role and permission profiles were reviewed in the application settings. However, complete role-based access testing with different user accounts and full email delivery testing were not performed.

The testing focused mainly on checking the configured features and correcting issues identified during development.

6.2 Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Add a new laboratory	Enter laboratory details and submit	Laboratory record created	Laboratory record saved successfully	Pass
TC-02	FR-01	Submit	Leave	Validation	Required	Pass

		laboratory details with missing fields	required fields empty and submit	message displayed	field validation checked	
TC-03	FR-02	Register new equipment	Enter equipment details and select laboratory	Equipment record created	Equipment record saved successfully	Pass
TC-04	FR-02	Select laboratory using lookup	Choose a laboratory in the equipment form	Laboratory linked to equipment	Lookup field checked successfully	Pass
TC-05	FR-03	Register a technician	Enter technician details and submit	Technician record created	Technician record saved successfully	Pass
TC-06	FR-04	Create calibration schedule	Enter schedule details and submit	Schedule record created	Schedule record saved successfully	Pass
TC-07	FR-04	Assign technician to calibration schedule	Select technician and submit	Technician assignment saved	Assignment configuration checked	Pass
TC-08	FR-05	Submit calibration record	Enter calibration details and result	Calibration record created	Calibration record saved successfully	Pass

TC-09	FR-05	Enter calibration parameters	Add parameter values in the subform	Parameter details stored	Subform configuration checked	Pass
TC-10	FR-06	Create calibration certificate	Enter certificate details and submit	Certificate record created	Certificate record saved successfully	Pass
TC-11	FR-07	Submit maintenance request	Enter maintenance details and submit	Maintenance request created	Maintenance request saved successfully	Pass
TC-12	FR-07	Record equipment inspection	Enter inspection details and submit	Inspection record created	Inspection record saved successfully	Pass
TC-13	FR-08	View reports and dashboard	Open reports and dashboard	Reports and dashboard displayed	Reports and dashboard displayed	Pass
TC-14	FR-09	Check overdue schedule status	Submit a schedule with a past date	Status changes to Overdue	Overdue status updated successfully	Pass
TC-15	FR-10	Check failed calibration alert	Review the failed calibration workflow configuration	Email alert configured for failed results	Workflow condition and email action configured	Pass

Table 6.1: Test Cases

6.3 Debugging Deluge Scripts

Debugging was performed during the development of CALIBTRACK to identify and correct errors in Deluge scripts. The script editor was used to review the code and check error messages displayed during execution. Field names, form link names, conditions and data types were checked whenever an error occurred.

One issue occurred in the Equipment Status on Failure workflow. The script displayed an error stating that the form 'Equipment' did not exist in the application. This indicated that the form name used in the script did not match the actual form link name. The issue was identified by reviewing the error message and checking the form name in Zoho Creator.

The debugging process helped identify configuration and scripting errors during development. Scripts were reviewed and tested after making corrections.

6.4 Defect Log

Defect ID	**Description**	**Root Cause**	Fix Applied	**Status**
BUG-01	Equipment Status on Failure workflow displayed an error stating that the Equipment form did not exist.	Incorrect form link name used in the Deluge script.	Identified the form-name mismatch. Correction and successful retesting were not confirmed.	Correction and successful retesting were not confirmed. Open

Table 6.2: Defect Log

6.5 User Acceptance Testing

User	Role Tested	Feedback	Action Taken
User 1	Laboratory Technician	The application is easy to use. Calibration records and equipment details can be entered easily.	Reviewed the form layout and field arrangement.
User 2	Equipment Manager	The dashboard and reports help monitor equipment, calibration schedules and maintenance requests.	Reviewed the dashboard layout and report presentation.

Table 6.3: User Acceptance Feedback

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

CALIBTRACK was developed and configured in the Development environment of Zoho Creator. The forms, reports, dashboard and workflows were created and reviewed during development. The application features were tested using sample records to check their functionality. A separate Stage or Production deployment was not completed. The application remains in the Development environment and has not been published as a production application.

S.No.	Checklist Item	Done
1.	Application features reviewed in Development	✓
2.	Sample/test data removed or isolated	
3.	Production roles and permissions configured	
4.	Production connections authorised	
5.	Schedules and workflows configured	✓
6.	Application published and URL verified	
7.	Version and release notes recorded	

Table 7.1: Deployment Checklist

7.2 Performance Optimisation

CALIBTRACK was developed using Zoho Creator's built-in forms, reports, workflows and dashboard features. The application was organised into separate forms to manage laboratory, equipment, technician, calibration and maintenance information. Reports were created to display relevant records, and the dashboard presents important information

through KPI cards and charts. These design choices help users access and review information in an organised manner. No formal performance benchmarking or before-and-after measurement was carried out.

Optimisation Applied	Component	Before	After	Improvement
Organised data into separate forms	Application structure	Information managed in separate modules	Information organised by category	Easier data management
Created relevant reports	Reports	Records accessed through individual forms	Records viewed through dedicated reports	Easier information retrieval
Added KPI cards and charts	Dashboard	No centralised summary	Important information displayed in one dashboard	Better data visibility

Table 7.2: Performance Optimisation Results

7.3 Security Features

CALIBTRACK includes role-based access configuration to organise user permissions. Different roles and permission profiles have been created to manage access to application features. The available permission profiles include Read, Technician Permission and Write. Roles include CEO, Laboratory Technician, Equipment Manager and Calibration Supervisor. These configurations provide a basic structure for managing user access.

Advanced security features such as OAuth connections, encrypted custom fields and IP restrictions were not separately implemented or tested.

Security Control	Where Applied	Threat Mitigated
Permission profiles	Zoho Creator Users – Permissions	Unauthorised access to application features
Role configuration	Zoho Creator Users – Roles	Improper role-based access
Read permission	Read permission profile	Unauthorised modification of records
Technician Permission	Technician Permission profile	Unnecessary access beyond assigned work

Table 7.3: Security Controls Implemented

7.4 Governance and Administration

CALIBTRACK was developed and maintained using Zoho Creator. The application forms, reports, workflows and dashboard were organised with clear names to make them easier to identify and manage. User roles and permission profiles were configured to organise application access. Deluge scripts and workflows were reviewed during development to identify errors. The application was maintained in the Development environment. Production deployment, backup procedures and formal change-management activities were not completed as part of this project.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The CALIBTRACK application was demonstrated using a laboratory equipment calibration scenario. The process starts with registering equipment and creating a calibration schedule. A technician's details are assigned to the schedule. After calibration, the result and parameter details are recorded. Certificate information is maintained for future reference. Finally, the dashboard and reports are used to view equipment, calibration and maintenance information.

Equipment ID	1
Equipment Name	Digital Weighing Machine
Equipment Category	Measuring Instrument
Laboratory	Physics Laboratory
Manufacturer	ExampleTech
Model Number	WM-100
Serial Number	SN1001
Purchase Date	15-Jan-2025
Purchase Cost	₹ 15,000.00
Last Calibration Date	15-Oct-2026
Calibration Interval	6 Months
Equipment Status	Active
Equipment Image	
Equipment Description	Digital weighing machine used for laboratory measurements and calibration testing.
Next Calibration Date	15-Apr-2027

Figure 8.1: Equipment Registration Details in CALIBTRACK

Schedule ID	1
Equipment	Digital Weighing Machine
Laboratory	Physics Laboratory
Calibration Type	Routine Calibration
Scheduled Date	05-Nov-2026
Assigned Technician	Arun Kumar
Priority	High
Schedule Status	Scheduled
Remarks	Test calibration schedule and technician assignment notification.

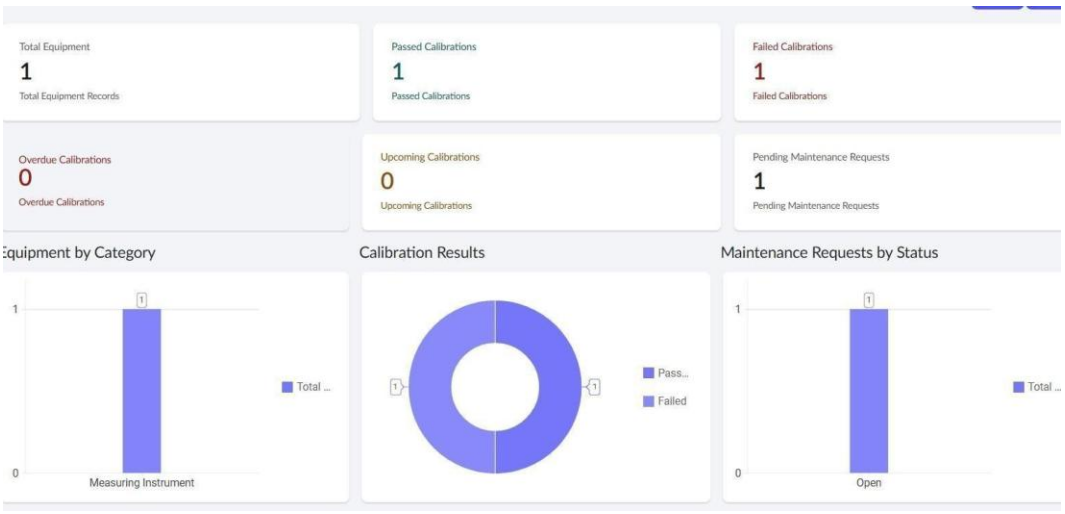
Figure 8.2: Calibration Schedule Details in CALIBTRACK

Calibration Record ID	1
Equipment	Digital Weighing Machine
Calibration Date	29-Oct-2026
Technician	Arun Kumar
Calibration Method	Routine Calibration
Calibration Result	Failed
Overall Remarks	Test failure alert workflow.
Next Calibration Date	29-Apr-2027
Calibration Parameters	Weight Accuracy
Unit	g
Result	Fail
Remarks	Actual value exceeds the allowed tolerance

Figure 8.3: Calibration Record Details and Parameter Results in CALIBTRACK

Certificate ID	CERT001
Certificate Number	CAL-2026-001
Equipment	Digital Weighing Machine
Calibration Record Form	1
Issue Date	29-Oct-2026
Valid From	29-Oct-2026
Valid Until	29-Oct-2027
Certificate Status	Valid
Certificate File	
Remarks	Test certificate submission and notification workflow.

Figure 8.4: Calibration Certificate Details in CALIBTRACK



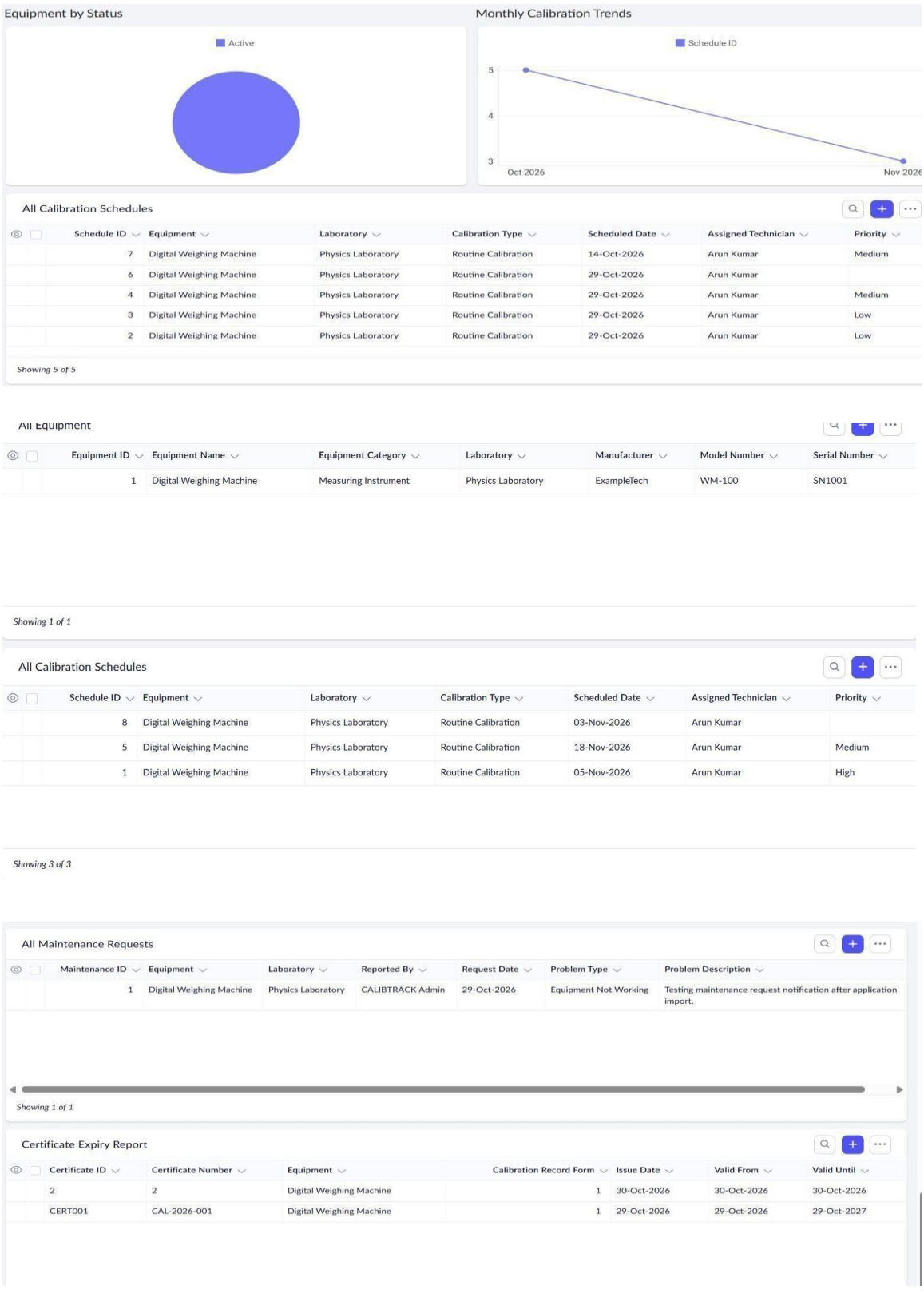


Figure 8.5 – Dashboard and Reports.

8.2 Achievement of Objectives

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To design and develop a laboratory equipment management application	Sections 5.1 and 5.2	CALIBTRACK application with eight forms and one subform
To create forms for maintaining laboratory, equipment and technician details	Section 5.2	Laboratory, Equipment and Technician forms.
To automate calibration scheduling and record management	Sections 5.2, 5.5 and 5.6	Calibration Schedule and Calibration Record forms with selected Deluge scripts and workflows
To generate calibration certificates and maintain certificate details	Section 5.2	Calibration Certificate form and certificate records
To develop reports and a dashboard for monitoring activities	Sections 5.3 and 5.4	15 reports, six KPI cards and five dashboard charts
To test the application features and verify their operation	Chapter 6	Test cases and confirmed overdue schedule status update

Table 8.1: Mapping of Objectives to Outcomes

8.3 Comparison with the Existing System

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Manual entry and tracking	Records entered through Zoho Creator forms	More organised processing
Manual steps	Separate records and manual follow-up	Forms and selected automated workflows	Reduced repetitive work

Report generation	Manual collection of information	15 reports available in the application	Easier report access
Data accessibility	Information maintained separately	Centralised records in CALIBTRACK	Easier information retrieval

Table 8.2: Existing System versus Proposed Application

8.4 Limitations of the Low-Code Approach

Zoho Creator provides useful low-code features for developing applications, but it also has some limitations. The application depends on the features and customisation options provided by the platform. Complex requirements may need additional Deluge scripting. External API integrations require proper connection configuration and authentication. The appearance of forms and reports is also limited by the available design options. CALIBTRACK was developed mainly using built-in forms, reports, workflows and selected Deluge scripts. Advanced integrations and custom extensions were not included in this project. These limitations were managed by using the available Zoho Creator features to meet the main project requirements.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

CALIBTRACK mainly follows the low-code development approach. The forms, reports, dashboard and workflows were created using Zoho Creator's visual development tools. Some automation requirements needed Deluge scripting, such as calculating the next calibration date, updating overdue schedule status and supporting email notifications. This represents the low-code approach because visual configuration was combined with programming.

The project did not use advanced pro-code extensions such as Zoho Catalyst, JavaScript widgets or external REST API integrations. If the application is expanded in the future, these technologies could be considered for more complex requirements. This project helped demonstrate how low-code development can be used to create a laboratory management application with limited programming.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The CALIBTRACK application was developed using Zoho Creator to manage laboratory equipment, calibration activities and maintenance records. The application includes eight main forms and one subform to store laboratory, equipment, technician, calibration schedule, calibration record, certificate, maintenance request and inspection details. It also includes 15 reports and a dashboard with KPI cards and charts to display important information.

Selected workflows and Deluge scripts were configured to support date calculations, calibration status updates and email notifications. The project provided practical experience in designing forms, managing data, creating reports, configuring workflows and testing a low-code application. Overall, CALIBTRACK demonstrates how Zoho Creator can be used to organise laboratory calibration activities in a single application.

9.2 Future Enhancements

1. Implement AI-based predictive analytics to identify equipment that may require calibration or maintenance.
2. Integrate external laboratory management systems through REST APIs.
3. Develop a mobile-friendly experience for technicians to update calibration records.
4. Add advanced role-based access testing and approval workflows.
5. Introduce automated certificate generation and downloadable PDF certificates.

9.3 Learning Outcomes and Career Relevance

Through this project, I gained practical experience in developing applications using Zoho Creator. I learned to create forms, configure lookup fields, design reports and build dashboards. I also gained experience in using Deluge scripts to perform date calculations and automate selected application activities. Working with workflows helped me understand how repetitive tasks and notifications can be automated.

This project improved my problem-solving, application design, testing and debugging skills. These skills are useful for career roles such as low-code application developer, citizen developer, software developer and automation engineer. The project also helped me understand how low-code platforms can be used to solve real-world management problems.

CHAPTER 10

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

No.	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and menu organised	Y	CALIBTRACK application	5.1	Fig. 5.1
II	Forms with 8 or more distinct field types	Y	Eight main forms	5.2	Figs. 5.2–5.19
II	Lookup fields between forms	Y	Equipment, Schedule, Record and Certificate forms	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	Y	Calibration Parameters subform	4.2, 5.2	Figs. 5.12–5.13
II	Field validations / layout rules	N	Not confirmed	5.2	—
II	List reports with filter, grouping and custom action	N	Reports created; all listed features not verified	5.3	Figs. 5.20–5.25

II	Kanban / Calendar / Timeline / Map report	N	Not implemented	5.3	—
II	Pivot table / chart report	N	Dashboard charts are used instead	5.4	Fig. 5.26
II	Page / dashboard built with UI Builder	Y	Calibration Dashboard	5.4	Fig. 5.26
II	Deluge – form events	Y	Calibration Date and Schedule Status workflows	5.5	DS-01, DS-02
II	Deluge – functions, fetch, loops, collections	Y	Technician lookup and certificate notification scripts	5.5	DS-03, DS-04
II	Email / in-app notifications	Y	Selected email notification workflows	5.5, 5.6	Figs. 5.27–5.33
II	Form workflow	Y	Form-based workflows	5.6	Figs. 5.27–5.33
II	Scheduled workflow	Y	Daily Calibration Status Check	5.6	Workflow configuration
II	Approval process	N	Not implemented	5.6	—
II	Blueprint	N	Not implemented	5.6	—

II	Batch / report workflow	N	Not implemented	5.6	—
II	Testing and debugging evidence	Y	Test cases and debugging discussion	6.2, 6.3	Table 6.1
III	Integration with a Zoho service	N	No separate service integration	5.7	—
III	Connection / REST API call (invokeurl)	N	Not implemented	5.7	—
III	Webhook or Creator REST API	N	Not implemented	5.7	—
III	Users, roles and permission profiles	Y	Users – Roles and Permissions	5.8	Fig. 5.34
III	Record-level access criteria / customer portal	N	Not implemented or verified	5.8, 5.11	—
III	Zia / AI feature	Y	Zia-assisted dashboard layout	5.9	Fig. 5.35
III	Analytics (charts / Zoho Analytics)	Y	Calibration Dashboard charts	5.9	Fig. 5.35
IV	Performance optimisation applied and measured	N	No formal performance measurement	7.2	Table 7.2
IV	Deployment across environments	N	Development environment only	7.1	—

IV	Security controls (encryption, OAuth, permissions)	N	Permission profiles configured; advanced controls not implemented	7.3	Fig. 5.34
IV	Governance (audit trail, administration)	N	Formal audit trail not implemented		—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not implemented	5.10	—
V	Low-code limitations and spectrum analysis	Y	Limitations and development reflection	8.4, 8.5	—
	Mobile app / portal access	N	Not configured or tested	5.11	—

Table A.1: Feature Coverage Checklist

APPENDIX B – COURSE OUTCOME MAPPING

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Chapters 1, 2 and 4	3 – High	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Chapters 4, 5 (Sections 5.1–5.6) and 6	3 – High	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Chapters 4, 5 (Sections 5.7–5.9)	2 – Medium	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Chapter 7	2 – Medium	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Chapter 5 (Sections 5.9, 5.10), Chapter 8 (Sections 8.4, 8.5) and Chapter 9 (Section 9.3)	3 – High	

Table B.1: Mapping of Course Outcomes to Report Chapters

APPENDIX C – COMPLETE DELUGE SOURCE CODE

fx Deluge script

```
if(input.Calibration_Date != null)
{
    input.Next_Calibration_Date = input.Calibration_Date.addMonth(6);
}

```

C.1 Calculate Next Calibration Date

fx Deluge script

```
if(input.Scheduled_Date < zoho.currentdate)
{
    input.Schedule_Status = "Overdue";
}

```

C.2 Calibration Schedule Status Check



```
technicianRecord = Technician_Form[ID == input.Assigned_Technician];
if(technicianRecord.count() > 0)
{
    technicianEmail = technicianRecord.Email;
    if(technicianEmail != null && technicianEmail != "")
    {
        sendmail
        [
            from :zoho.adminuserid
            to :technicianEmail
            subject :"New Calibration Task Assigned - CALIBTRACK"
            message :"Dear Technician,<br><br>A new calibration task has been assigned to you. Please log in to CALIBTRACK and check your calibration schedule.<br><br>Thank you,<br>CALIBTRACK System"
        ]
    }
}
```

C.3 Technician Assignment Notification



```
if(input.Valid_Until <= zoho.currentdate.addDay(7))
{
    sendmail
    [
        from :zoho.loginuserid
        to :"dharshini.g2.2024.cse@rajalakshmi.edu.in"
        subject :"CALIBTRACK - Certificate Expiry Alert"
        message :"Dear Admin,\n\nA calibration certificate is expiring soon.\n\nCertificate ID: " + input.Certificate_ID + "\nCertificate Number: " + input.Certificate_Number + "\nValid Until: " + input.Valid_Until + "\n\nPlea
arrange certificate renewal.\n\nRegards,\nCALIBTRACK System"
    ]
}
```

C.4 Certificate Submission Notification

APPENDIX D – USER MANUAL

D,1 Accessing the Application

1. Open Google Chrome or another supported web browser.
2. Open the CALIBTRACK application using its Zoho Creator application link.
3. Log in using your authorised Zoho Creator account.
4. After successful login, the CALIBTRACK home page will be displayed.
5. Use the left-side navigation menu to access forms, reports and the Calibration Dashboard.

Note: Mobile application access was not separately configured or tested in this project.

D.2 Instructions for Laboratory Technician

1. Log in to the CALIBTRACK application.
2. Open the Calibration Schedules section.
3. View the assigned equipment and calibration schedule.
4. Open the Calibration Record form.
5. Enter the calibration date, method and calibration result.
6. Enter the calibration parameter details in the subform.
7. Submit the calibration record.
8. View the saved record in the Calibration History Report.

D.3 Instructions for Equipment Manager

1. Log in to the CALIBTRACK application.
2. Open the Equipment section.
3. Add new equipment details or view existing equipment records.
4. Open the Maintenance Requests section to review maintenance requests.
5. Create a new maintenance request when equipment requires repair.
6. Open the Equipment Inspection section to record inspection details.
7. Use the Equipment Report and Maintenance Report to review the available information.
8. Open the Calibration Dashboard to view equipment and maintenance summaries.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	[ ]	Register No.	[ ]
Project Title	[ ]	Date	[ ]

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports& Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC& AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Table E.2: Performance Level Descriptors

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	[ ]	Register No.	[ ]
Department/ Section	[ ]	Date of Viva	[ ]
Application URL Verified	Yes / No	Demo Duration	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____